Ohio | Department of Education

# Ohio's State Tests

## ITEM RELEASE

## SPRING 2023

## GEOMETRY



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# Geometry
# Spring 2023 Item Release
# Content Summary and Answer Key

| Question No.* | Item Type | Content Cluster | Content Standard | Depth of Knowledge | Answer Key | Points |
|---|---|---|---|---|---|---|
| 1 | Multiple Choice Item | Find arc lengths and areas of sectors of circles. | Find arc lengths and areas of sectors of circles.<br>*b.* Derive the formula for the area of a sector, and use it to solve problems. *(G.C.5)* | Level 1 | A | 1 point |
| 2 | Matching Item | Apply geometric concepts in modeling situations. | Use geometric shapes, their measures, and their properties to describe objects, e.g., modeling a tree trunk or a human torso as a cylinder. *(G.MG.1)* | Level 1 | --- | 1 point |
| 5 | Equation Item | Understand indepen-dence and conditional probability, and use them to interpret data. | Understand that two events A and B are independent if and only if the probability of A and B occurring together is the product of their probabilities, and use this characterization to determine if they are independent. *(S.CP.2)* | Level 1 | --- | 1 point |
| 6 | Equation Item | Use coordinates to prove simple geometric theorems algebraically and to verify specific geometric statements. | Justify the slope criteria for parallel and perpendicular lines, and use them to solve geometric problems, e.g., find the equation of a line parallel or perpendicular to a given line that passes through a given point. *(G.GPE.5)* | Level 3 | --- | 2 points |
| 7 | Equation Item | Explain volume formulas and use them to solve problems. | Use volume formulas for cylinders, pyramids, cones, and spheres to solve problems. *(G.GMD.3)* | Level 2 | --- | 1 point |

** The question number matches the item number in the Item Level Report in the Centralized Reporting System. The items are numbered sequentially in the practice site.*

# Geometry
# Spring 2023 Item Release
# Content Summary and Answer Key

| Question No.* | Item Type | Content Cluster | Content Standard | Depth of Knowledge | Answer Key | Points |
|---|---|---|---|---|---|---|
| 9 | Equation Item | Understand similarity in terms of similarity transform-ations. | Verify experimentally the properties of dilations given by a center and a scale factor:<br>*a*. A dilation takes a line not passing through the center of the dilation to a parallel line and leaves a line passing through the center unchanged. *(G.SRT.1)* | Level 2 | --- | 1 point |
| 11 | Equation Item | Understand congruence in terms of rigid motions. | Use geometric descriptions of rigid motions to transform figures and to predict the effect of a given rigid motion on a given figure; given two figures, use the definition of congruence in terms of rigid motions to decide if they are congruent. *(G.CO.6)* | Level 2 | --- | 1 point |
| 16 | Multiple Choice Item | Experiment with transform-ations in the plane. | Know precise definitions of ray, angle, circle, perpendicular line, parallel line, and line segment, based on the undefined notions of point, line, distance along a line, and arc length. *(G.CO.1)* | Level 1 | B | 1 point |
| 18 | Multiple Choice Item | Experiment with transform-ations in the plane. | Given a geometric figure and a rotation, reflection, or translation, draw the transformed figure using items such as graph paper, tracing paper, or geometry software. Specify a sequence of transformations that will carry a given figure onto another. *(G.CO.5)* | Level 2 | B | 1 point |
| 27 | Equation Item | Apply geometric concepts in modeling situations. | Apply concepts of density based on area and volume in modeling situations, e.g., persons per square mile, BTUs per cubic foot.★ *(G.MG.2)* | Level 1 | --- | 1 point |

** The question number matches the item number in the Item Level Report in the Centralized Reporting System. The items are numbered sequentially in the practice site.*

# Geometry
# Spring 2023 Item Release
# Content Summary and Answer Key

| Question No.* | Item Type | Content Cluster | Content Standard | Depth of Knowledge | Answer Key | Points |
|---|---|---|---|---|---|---|
| 28 | Multiple Choice Item | Understand congruence in terms of rigid motions. | Use the definition of congruence in terms of rigid motions to show that two triangles are congruent if and only if corresponding pairs of sides and corresponding pairs of angles are congruent. *(G.CO.7)* | Level 2 | D | 1 point |
| 29 | Equation Item | Define trigonometric ratios and solve problems involving right triangles. | Solve problems involving right triangles.★ *a.* Use trigonometric ratios and the Pythagorean Theorem to solve right triangles in applied problems if one of the two acute angles and a side length is given. *(G.SRT.8)* | Level 3 | --- | 1 point |
| 30 | Multiple Choice Item | Prove geometric theorems both formally and informally using a variety of methods. | Prove and apply theorems about triangles. *Theorems include but are not restricted to the following: measures of interior angles of a triangle sum to 180°; base angles of isosceles triangles are congruent; the segment joining midpoints of two sides of a triangle is parallel to the third side and half the length; the medians of a triangle meet at a point. (G.CO.10)* | Level 2 | D | 1 point |
| 31 | Equation Item | Understand independ-ence and conditional probability, and use them to interpret data. | Understand that two events A and B are independent if and only if the probability of A and B occurring together is the product of their probabilities, and use this characterization to determine if they are independent.★ *(S.CP.2)* | Level 2 | --- | 1 point |

** The question number matches the item number in the Item Level Report in the Centralized Reporting System. The items are numbered sequentially in the practice site.*

# Geometry
# Spring 2023 Item Release
# Content Summary and Answer Key

| Question No.* | Item Type | Content Cluster | Content Standard | Depth of Knowledge | Answer Key | Points |
|---|---|---|---|---|---|---|
| 36 | Equation Item | Understand independ-ence and conditional probability, and use them to interpret data. | Construct and interpret two-way frequency tables of data when two categories are associated with each object being classified. Use the two-way table as a sample space to decide if events are independent and to approximate conditional probabilities. *For example, collect data from a random sample of students in your school on their favorite subject among math, science, and English. Estimate the probability that a randomly selected student from your school will favor science given that the student is in tenth grade. Do the same for other subjects and compare the results.★ (S.CP.4)* | Level 2 | --- | 2 points |
| 37 | Matching Item | Classify and analyze geometric figures. | Classify two-dimensional figures in a hierarchy based on properties. *(G.CO.14)* | Level 2 | --- | 1 point |
| 46 | Equation Item | Understand and apply theorems about circles. | Construct the inscribed and circumscribed circles of a triangle; prove and apply the property that opposite angles are supplementary for a quadrilateral inscribed in a circle. *(G.C.3)* | Level 1 | --- | 1 point |

** The question number matches the item number in the Item Level Report in the Centralized Reporting System. The items are numbered sequentially in the practice site.*

# Geometry
# Spring 2023 Item Release
# Content Summary and Answer Key

| Question No.* | Item Type | Content Cluster | Content Standard | Depth of Knowledge | Answer Key | Points |
|---|---|---|---|---|---|---|
| 47 | Multiple Choice Item | Use coordinates to prove simple geometric theorems algebraically and to verify specific geometric statements. | Justify the slope criteria for parallel and perpendicular lines, and use them to solve geometric problems, e.g., find the equation of a line parallel or perpendicular to a given line that passes through a given point. *(G.GPE.5)* | Level 1 | A | 1 point |
| 48 | Inline Choice Item | Understand similarity in terms of similarity transform-ations. | Given two figures, use the definition of similarity in terms of similarity transformations to decide if they are similar; explain using similarity transformations the meaning of similarity for triangles as the equality of all corresponding pairs of angles and the proportionality of all corresponding pairs of sides. *(G.SRT.2)* | Level 2 | --- | 1 point |
| 49 | Multiple Choice Item | Understand independ-ence and conditional probability, and use them to interpret data. | Recognize and explain the concepts of conditional probability and independence in everyday language and everyday situations. *For example, compare the chance of having lung cancer if you are a smoker with the chance of being a smoker if you have lung cancer.★ (S.CP.5)* | Level 2 | D | 1 point |

** The question number matches the item number in the Item Level Report in the Centralized Reporting System. The items are numbered sequentially in the practice site.*

# Depth of Knowledge (DOK)

DOK refers to the complexity of thinking required to complete a task in a given item. Items with a DOK 1 designation focus on the recall of information, such as definitions and terms, and simple procedures. Items with a DOK 2 designation require students to make decisions, solve routine problems, perform calculations, or recognize patterns. Items with a DOK 3 designation feature higher-order cognitive tasks. These DOK 3 tasks include but are not limited to: critiquing a statement and forming a conclusion; explaining, justifying, or proving a statement; or approaching abstract, complex, open-ended, and non-routine problems. Each grade's blueprint contains information about the number of points of opportunity students will encounter at each DOK level.

**Table 1: *Math Descriptors* – Applying Depth of Knowledge Levels for Mathematics (Webb, 2002) & NAEP 2002 Mathematics Levels of Complexity**
(M. Petit, Center for Assessment 2003, K. Hess, Center for Assessment, updated 2006)

| Level 1<br>Recall | Level 2<br>Skills/Concepts | Level 3<br>Strategic Thinking | Level 4*<br>Extended Thinking |
|---|---|---|---|
| a. Recall, observe, or recognize a fact, definition, term, or property<br>b. Apply/compute a well-known algorithm (e.g., sum, quotient)<br>c. Apply a formula<br>d. Determine the area or perimeter of rectangles or triangles given a drawing and labels<br>e. Identify a plane or three-dimensional figure<br>f. Measure<br>g. Perform a specified or routine procedure (e.g., apply rules for rounding)<br>h. Evaluate an expression<br>i. Solve a one-step word problem<br>j. Retrieve information from a table or graph | a. Classify plane and three-dimensional figures<br>b. Interpret information from a simple graph<br>c. Use models to represent mathematical concepts<br>d. Solve a routine problem requiring multiple steps/ decision points, or the application of multiple concepts<br>e. Compare and/or contrast figures or statements<br>f. Construct 2-dimensional patterns for 3-dimensional models, such as cylinders and cones<br>g. Provide justifications for steps in a solution process<br>h. Extend a pattern | a. Interpret information from a complex graph<br>b. Explain thinking when more than one response is possible<br>c. Make and/or justify conjectures<br>d. Use evidence to develop logical arguments for a concept<br>e. Use concepts to solve non-routine problems<br>f. Perform a procedure with multiple steps and multiple decision points<br>g. Generalize a pattern<br>h. Describe, compare, and contrast solution methods<br>i. Formulate a mathematical model for a complex situation<br>j. Provide mathematical justifications | a. Relate mathematical concepts to other content areas<br>b. Relate mathematical concepts to real-world applications in new situations<br>c. Apply a mathematical model to illuminate a problem, situation<br>d. Conduct a project that specifies a problem, identifies solution paths, solves the problem, and reports results<br>e. Design a mathematical model to inform and solve a practical or abstract situation<br>f. Develop generalizations of the results obtained and the strategies used and apply them to new problem situations |

**Table 1** *continued on next page.*

| Level 1<br>Recall | Level 2<br>Skills/Concepts | Level 3<br>Strategic Thinking | Level 4*<br>Extended Thinking |
|---|---|---|---|
| k. Recall, identify, or make conversions between and among representations or numbers (fractions, decimals, and percents), or within and between customary and metric measures<br>l. Locate numbers on a number line, or points on a coordinate grid<br>m. Solve linear equations<br>n. Represent math relationships in words, pictures, or symbols<br>o. Read, write, and compare decimals in scientific notation | i. Retrieve information from a table, graph, or figure and use it to solve a problem requiring multiple steps<br>j. Translate between tables, graphs, words and symbolic notation<br>k. Make direct translations between problem situations and symbolic notation<br>l. Select a procedure according to criteria and perform it<br>m. Specify and explain relationships between facts, terms, properties, or operations<br>n. Compare, classify, organize, estimate, or order data | k. Solve a multiple-step problem and provide support with a mathematical explanation that justifies the answer<br>l. Solve 2-step linear equations/ inequalities in one variable over the rational numbers, interpret solution(s) in the original context, and verify reasonableness of results<br>m. Translate between a problem situation and symbolic notation that is not a direct translation<br>n. Formulate an original problem, given a situation<br>o. Analyze the similarities and differences between procedures<br>p. Draw conclusion from observations or data, citing evidence | g. Apply one approach among many to solve problems<br>h. Apply understanding in a novel way, providing an argument/ justification for the application<br><br>***NOTE:*** *Level 4 involves such things as complex restructuring of data or establishing and evaluating criteria to solve problems.* |

****Note: Ohio's State Tests only assess and measure DOK Levels 1 – 3 in grades K – 12. Level 4 is included in this table for informational purposes only.***

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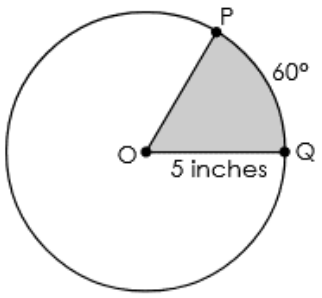
# Geometry
# Spring 2023 Item Release

# Question 1

# Question and Scoring Guidelines

# Question 1

Circle O is shown, where $m\overset{\frown}{PQ} = 60°$ and $OQ = 5$ inches.




Which expression represents the area, in square inches, of the shaded sector?

Ⓐ $\frac{60}{360}\pi(5)^2$

Ⓑ $\frac{60}{360}\pi(10)^2$

Ⓒ $\frac{60}{180}\pi(5)^2$

Ⓓ $\frac{60}{180}\pi(10)^2$

**Points Possible:** 1

**Content Cluster:** Find arc lengths and areas of sectors of circles.

**Content Standard:** Find arc lengths and areas of sectors of circles. *b.* Derive the formula for the area of a sector, and use it to solve problems. *(G.C.5)*

**Depth of Knowledge:** Level 1

**Modeling and Reasoning:** No

# Scoring Guidelines

Rationale for Option A: **Key** – The student identifies that the area of the sector is proportional to the angular measure of the arc bounded by this sector. Since the angular measure of an arc is 60°, which is $\frac{60}{360}$ part of the entire angular measure of a circle, the area of a sector is the same part $\frac{60}{360}$, of the area of a circle, which is $\frac{60}{360} * \pi(5)^2$.

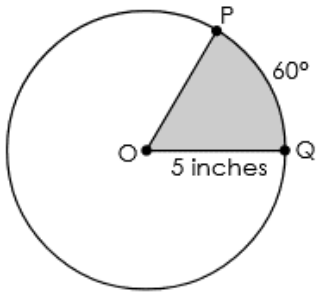
Rationale for Option B: This is incorrect. The student may identify that the area of the sector is proportional to the angular measure of the arc bounded by this sector. Meaning, since the angular measure of an arc is 60°, which is $\frac{60}{360}$ part of the entire angular measure of a circle, the area of a sector is the same part, $\frac{60}{360}$ of the area of a circle. However, the student may identify an expression that involves the diameter (10), instead of the radius (5).

Rationale for Option C: This is incorrect. The student may understand that the sector is a fraction of the circle's area and chooses an expression that includes the area of the circle, $\pi(5)^2$. However, the student may incorrectly think that the angular measure of a circle is 180°, the angular measure for the arc is $\frac{60}{180}$ part of the angular measure of a circle and the area of a sector is $\frac{60}{180} * \pi(5)^2$.

Rationale for Option D: This is incorrect. The student may think that the angular measure of a circle is 180° and the angular measure for the arc is $\frac{60}{180}$ part of an angular measure of the circle. Also, the student may understand that the area of the sector is a fraction of the area of the circle, but identifies an expression for the area of the circle that involves the diameter (10), instead of the radius (5).

## Sample Response: 1 point

Circle O is shown, where $m\overset{\frown}{PQ} = 60°$ and $OQ = 5$ inches.




Which expression represents the area, in square inches, of the shaded sector?

- ● $\frac{60}{360}\pi(5)^2$
- Ⓑ $\frac{60}{360}\pi(10)^2$
- Ⓒ $\frac{60}{180}\pi(5)^2$
- Ⓓ $\frac{60}{180}\pi(10)^2$

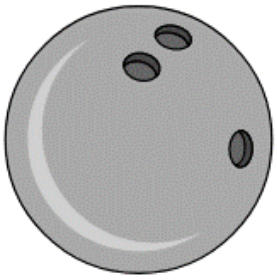
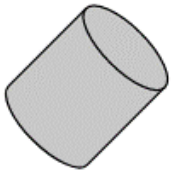
# Geometry
# Spring 2023 Item Release

# Question 2

# Question and Scoring Guidelines

# Question 2

Match each real-life object to the geometric figure that **best** models it.

| | Cylinder | Cone | Sphere |
|---|---|---|---|
| **Bowling Ball**  | ☐ | ☐ | ☐ |
| **Pencil Eraser**  | ☐ | ☐ | ☐ |
| **Melon**  | ☐ | ☐ | ☐ |

**Points Possible:** 1

**Content Cluster:** Apply geometric concepts in modeling situations.

**Content Standard:** Use geometric shapes, their measures, and their properties to describe objects, e.g., modeling a tree trunk or a human torso as a cylinder. *(G.MG.1)*

**Depth of Knowledge:** Level 1

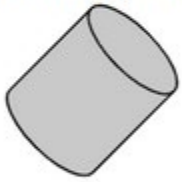
**Modeling and Reasoning:** Yes

# Scoring Guidelines

For full credit (1 point), the student's response satisfies the bullet below.

- The student selects the correct answers, providing evidence of the ability to use geometric shapes and their properties to describe objects.

Exemplar Response

| | Cylinder | Cone | Sphere |
|---|---|---|---|
| **Bowling Ball**  | ☐ | ☐ | ☑ |
| **Pencil Eraser**  | ☑ | ☐ | ☐ |
| **Melon**  | ☐ | ☐ | ☑ |

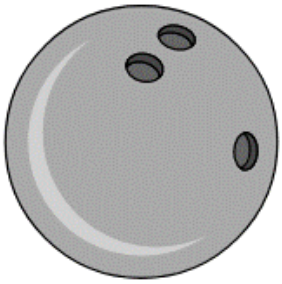
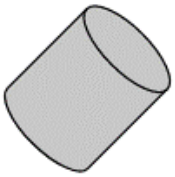
# Geometry
# Spring 2023 Item Release

# Question 2

# Sample Responses

## Sample Response: 1 point

Match each real-life object to the geometric figure that **best** models it.

| | Cylinder | Cone | Sphere |
|---|---|---|---|
| **Bowling Ball**  | ☐ | ☐ | ☑ |
| **Pencil Eraser**  | ☑ | ☐ | ☐ |
| **Melon**  | ☐ | ☐ | ☑ |

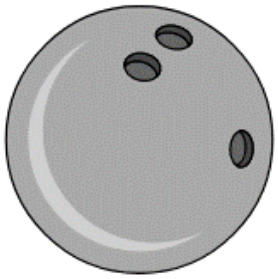
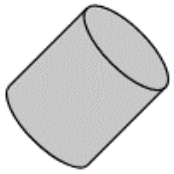
**Notes on Scoring**

This response earns full credit (1 point) because the student correctly matches each real-life object to the geometric figure that best models it.

The student identifies the bowling ball as a sphere, the pencil eraser as a cylinder and the melon as a sphere, matching each real-life object to the best geometric figure.

## Sample Response: 0 points

Match each real-life object to the geometric figure that **best** models it.

| | Cylinder | Cone | Sphere |
|---|---|---|---|
| **Bowling Ball**  | ☑ | ☐ | ☐ |
| **Pencil Eraser**  | ☐ | ☐ | ☑ |
| **Melon**  | ☑ | ☐ | ☐ |

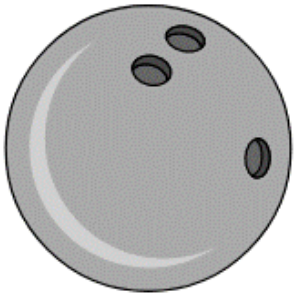
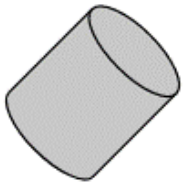
**Notes on Scoring**

This response earns no credit (0 points) because the student incorrectly matches each real-life object to the geometric figure that best models it.

The student may confuse a sphere and a cylinder, identifying the bowling ball as a cylinder when they mean a sphere, the pencil eraser as a sphere when they mean a cylinder, and the melon as a cylinder when they mean a sphere.

## Sample Response: 0 points

Match each real-life object to the geometric figure that **best** models it.

| | Cylinder | Cone | Sphere |
|---|---|---|---|
| **Bowling Ball**  | ☐ | ☐ | ☑ |
| **Pencil Eraser**  | ☑ | ☐ | ☐ |
| **Melon**  | ☐ | ☑ | ☐ |

**Notes on Scoring**

This response earns no credit (0 points) because the student incorrectly matches one of the real-life objects to the geometric figure that best models it.

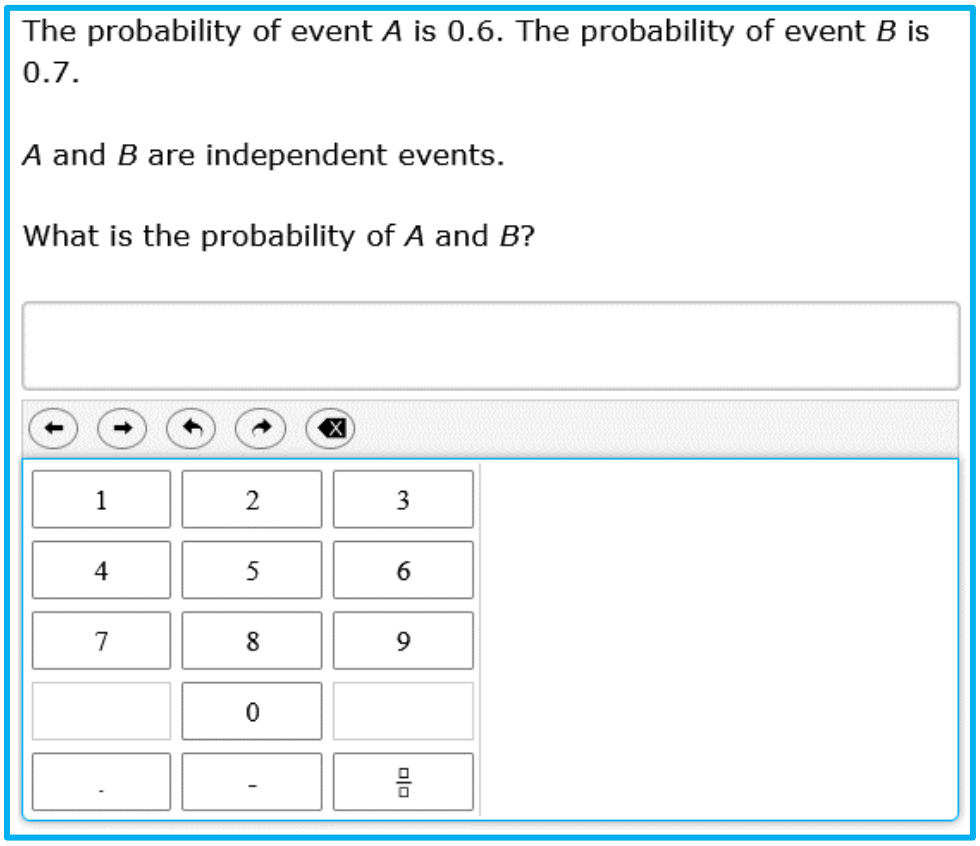
The student identifies the bowling ball as a sphere and the pencil eraser as a cylinder, matching the first two real-life objects to the best geometric figure. However, the student may try to incorporate the small stem of the melon into its corresponding geometric figure and think that the melon is best represented by a cone. The student does not recognize that the sphere is still the best model for the melon.

# Geometry
# Spring 2023 Item Release

# Question 5

# Question and Scoring Guidelines

# Question 5




**Points Possible:** 1

**Content Cluster:** Understand independence and conditional probability, and use them to interpret data.

**Content Standard:** Understand that two events A and B are independent if and only if the probability of A and B occurring together is the product of their probabilities, and use this characterization to determine if they are independent. *(S.CP.2)*

**Depth of Knowledge:** Level 1

**Modeling and Reasoning:** No

# Scoring Guidelines

For full credit (1 point), the student's response satisfies the bullet below.

- The student calculates the correct compound probability, providing evidence of the ability to understand that the two events are independent if the probability of them occurring together is the product of their probabilities.

Exemplar Response

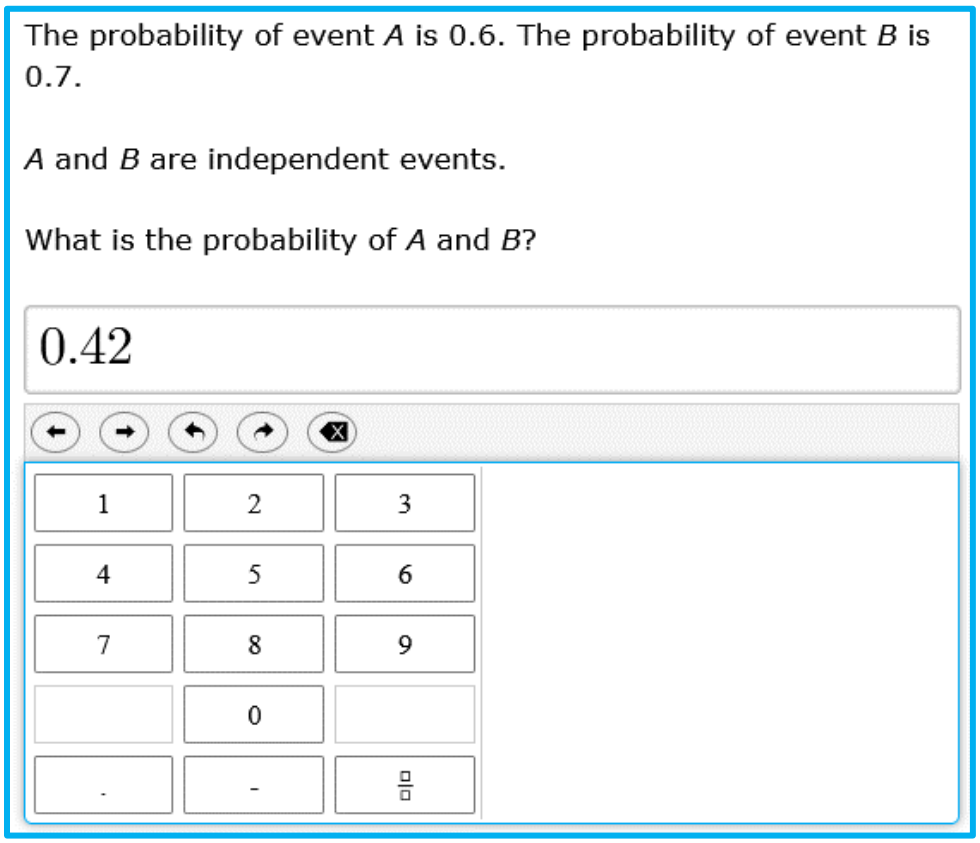
- 0.42
  or
- any equivalent value

# Geometry
# Spring 2023 Item Release

## Question 5

## Sample Responses

## Sample Response: 1 point




**Notes on Scoring**

This response earns full credit (1 point) because the student correctly identifies the probability of *A* and *B* given that they are independent events.

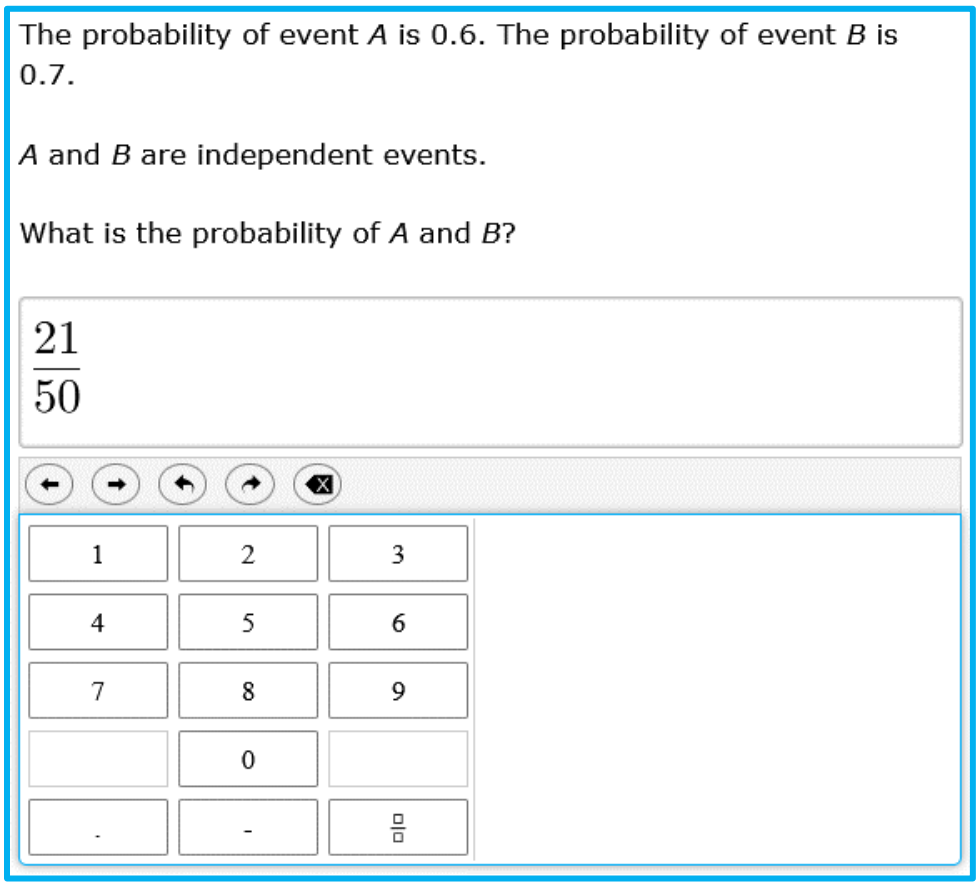
The student understands that since event *A* and event *B* are independent, the probability of *A* and *B* occurring, *P*(*A* and *B*), is equal to the product of their individual probabilities.

$$P(A \text{ and } B) = P(A) \cdot P(B)$$

$$= 0.6 \cdot 0.7$$

$$= 0.42$$

## Sample Response: 1 point



**Notes on Scoring**

This response earns full credit (1 point) because the student correctly identifies the probability of *A* and *B* given, they are independent events.

The student understands that since event *A* and event *B* are independent, the probability of *A* and *B* occurring, *P*(*A* and *B*), is equal to the product of their probabilities. The student may change the decimals to fractions when calculating the probability.

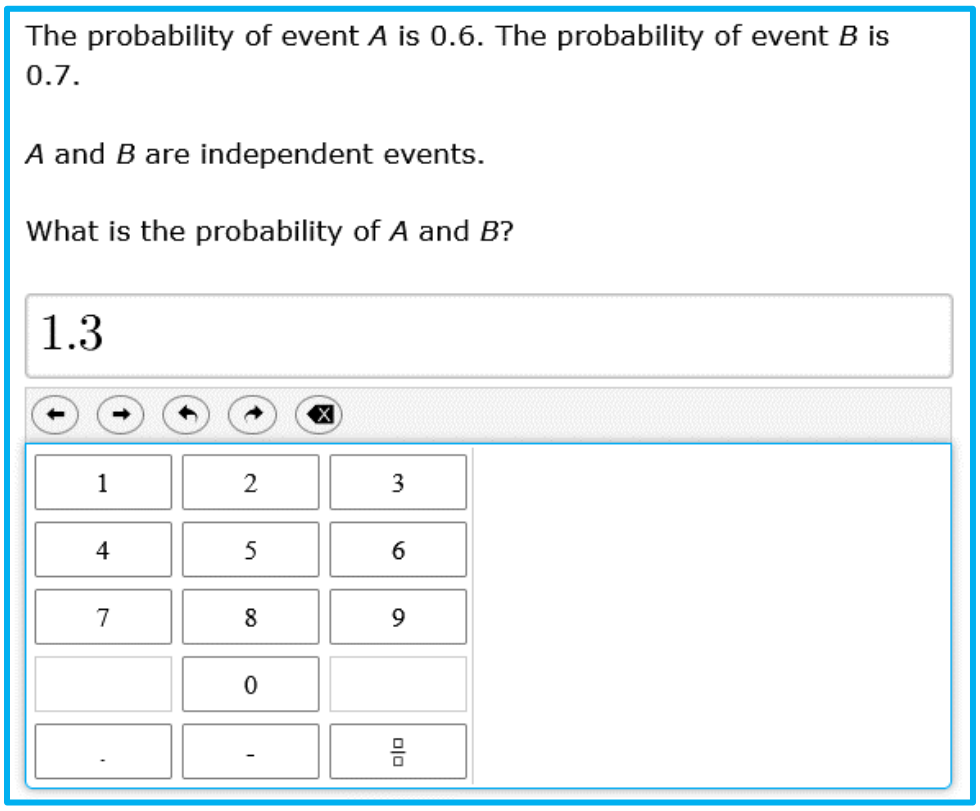
$$P(A \text{ and } B) = P(A) \cdot P(B)$$

$$= \frac{6}{10} \cdot \frac{7}{10}$$

$$= \frac{42}{100}$$

$$= \frac{21}{50}$$

## Sample Response: 0 points



**Notes on Scoring**

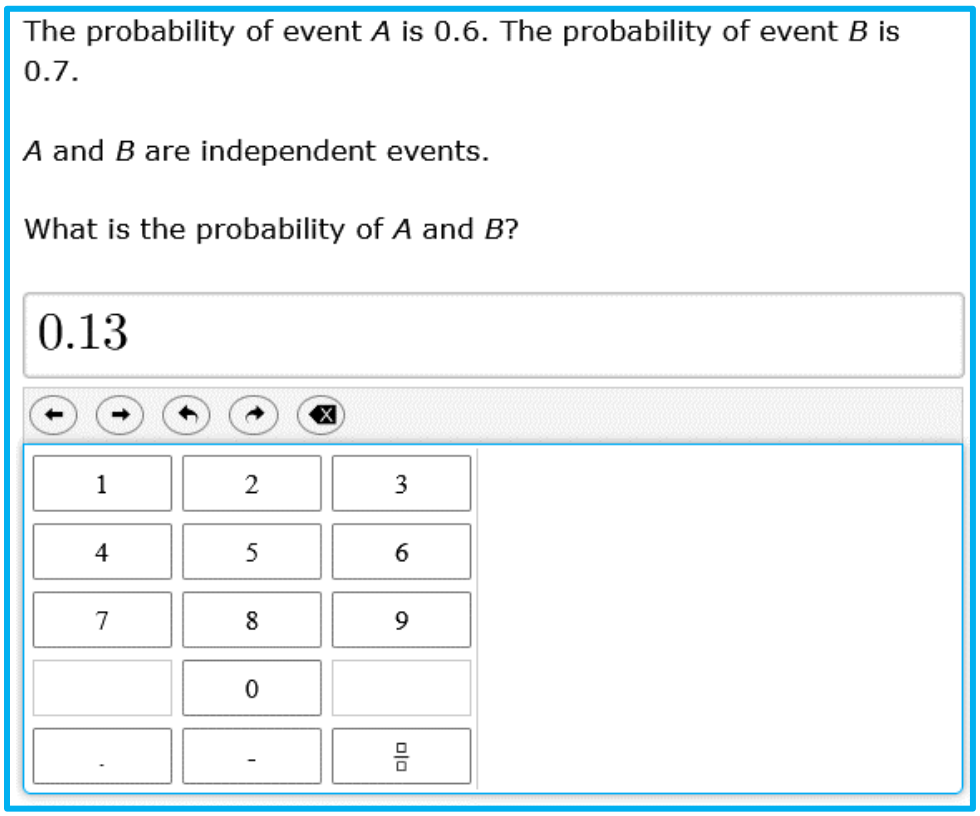
This response earns no credit (0 points) because the student incorrectly identifies the probability of *A* and *B* given, they are independent events.

The student may think that since event *A* and event *B* are independent, the probability of *A* and *B* occurring, *P*(*A* and *B*), is equal to the sum of the individual probabilities instead of the product.

$$\begin{aligned} P(A \text{ and } B) &= P(A) + P(B) \\ &= 0.6 + 0.7 \\ &= 1.3 \end{aligned}$$

The student also does not recognize that 1.3 is not a reasonable response since the probability of any event must be a value from 0 to 1.

## Sample Response: 0 points



**Notes on Scoring**

This response earns no credit (0 points) because the student incorrectly identifies the probability of *A* and *B* given, they are independent events.

The student may think since event A and event B are independent, the probability of A and B occurring, *P*(*A* and *B*), is equal to the sum of their individual probabilities instead of the product. The student may confuse the rules for where to place the decimal when multiplying or adding decimals, and places two digits to the right of the decimal point in the sum since there is a total of two digits to the right of the decimals in 0.6 and 0.7.

$$
\begin{aligned}
P(A \text{ and } B) &= P(A) + P(B) \\
&= 0.6 + 0.7 \\
&= \ 6 \ + \ 7 \\
&= 13 \\
&= .13
\end{aligned}
$$

# Geometry
# Spring 2023 Item Release

# Question 6

# Question and Scoring Guidelines

# Question 6

Triangle ABC, with the following characteristics, is placed on the coordinate plane.

- $\overline{AB}$ is on a vertical line.
- $\angle C$ is a right angle.
- Point C is located at (3, 2).
- The slope of $\overline{AC}$ is 5.

What are possible coordinates for points A and B?

A ( , )

B ( , )



**Points Possible:** 2

**Content Cluster:** Use coordinates to prove simple geometric theorems algebraically and to verify specific geometric statements.

**Content Standard:** Justify the slope criteria for parallel and perpendicular lines, and use them to solve geometric problems, e.g., find the equation of a line parallel or perpendicular to a given line that passes through a given point. *(G.GPE.5)*

**Depth of Knowledge:** Level 3

**Modeling and Reasoning:** Yes

# Scoring Guidelines

For 2 points, the student's response satisfies both bullets below.

For 1 point, the student's response satisfies one of the bullets below.

- The student correctly creates two points with the correct slopes in relation to point C, providing evidence of an ability to identify the slope of a perpendicular line with a nonzero slope.
- The student correctly creates 2 unique points on a vertical line, providing evidence of an ability to create points in relation to a geometric construction.

Exemplar Response: 2 points

- A (5, 12) and B (5, 1.6)

Exemplar Response: 1 point

- A (1, –8) and B (–7, 4)

# Geometry
# Spring 2023 Item Release

## Question 6

## Sample Responses

## Sample Response: 2 points

Triangle ABC, with the following characteristics, is placed on the coordinate plane.

- $\overline{AB}$ is on a vertical line.
- $\angle C$ is a right angle.
- Point C is located at (3, 2).
- The slope of $\overline{AC}$ is 5.

What are possible coordinates for points A and B?

A ( 5 , 12 )

B ( 5 , 1.6 )



**Notes on Scoring**

This response earns full credit (2 points) because the student finds correct coordinates for point A and point B.

The student finds the coordinates for point A by graphing point C at (3, 2) and then using the given information that the slope of $\overline{\mathrm{AC}}$ is 5. Since slope is defined as the change in *y* divided by the change in x, a slope of 5 represents the change of 5 in the *y*-direction and a change of 1 in the x-direction. This pair of changes is applied twice from point C, moving up 10 units and right 2 units to identify a possible location for point A at (5, 12).

Since $\overline{\mathrm{AB}}$ is on a vertical line, points A and B have the same x-coordinate, 5. To find the *y*-coordinate for point B, the student recalls the given information that angle C is a right angle, or that $\overline{\mathrm{AC}}$ and $\overline{\mathrm{BC}}$ are perpendicular, which means the product of their slopes is $-1$. Since the slope of $\overline{\mathrm{AC}}$ is 5, the slope of $\overline{\mathrm{BC}}$ must be $-\frac{1}{5}$. The slope formula can now be used to calculate the *y*-coordinate of point B using the coordinates for point C as the other point in the formula.

$$\frac{2-y}{(3-5)} = -\frac{1}{5}$$

$$\frac{2-y}{(-2)} = -\frac{1}{5}$$

$$2-y = \frac{2}{5}$$

$$-y = -\frac{8}{5}$$

$$y = \frac{8}{5}$$

The student writes the fraction as the decimal 1.6, thus, showing point B is located at (5, 1.6).

## Sample Response: 2 points

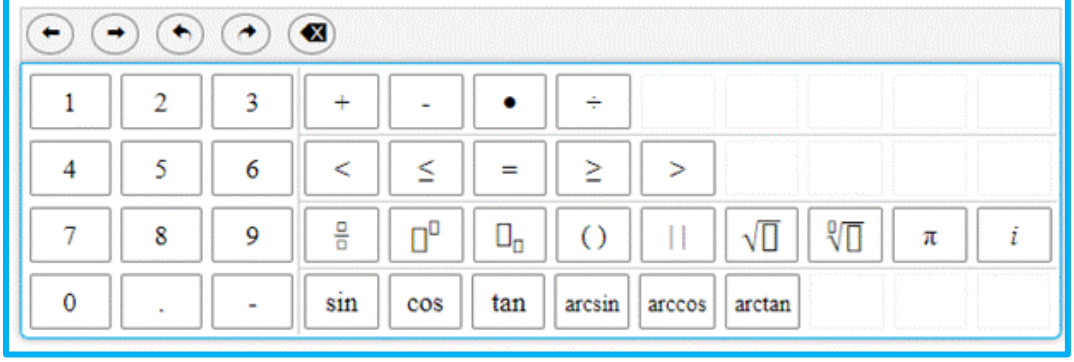
Triangle ABC, with the following characteristics, is placed on the coordinate plane.

- $\overline{AB}$ is on a vertical line.
- $\angle C$ is a right angle.
- Point C is located at (3, 2).
- The slope of $\overline{AC}$ is 5.

What are possible coordinates for points A and B?

A ( 1 , −8 )

B ( 1 , 2.4 )



**Notes on Scoring**

This response earns full credit (2 points) because the student finds correct coordinates for point A and point B.

The student finds the coordinates for point A by graphing point C at (3, 2) and then using the given information that the slope of $\overline{\mathrm{AC}}$ is 5. Since slope is defined as the change in *y* divided by the change in x, a slope of 5 represents the change of 5 in the *y*-direction and a change of 1 in the x-direction. This pair of changes is applied twice from point C, moving down 10 units and left 2 units to identify a possible location for point A at (1, –8).

Since $\overline{\mathrm{AB}}$ is on a vertical line, points A and B have the same x-coordinate, 1. To find the *y*-coordinate for point B, the student uses the information that angle C is a right angle, or that $\overline{\mathrm{AC}}$ and $\overline{\mathrm{BC}}$ are perpendicular, which means the product of their slopes is $-1$. Since the slope of $\overline{\mathrm{AC}}$ is 5, the slope of $\overline{\mathrm{BC}}$ must be $-\frac{1}{5}$. The student now uses the slope formula to calculate the *y*-coordinate of point B using the coordinates for point C as the other point in the formula.

$$\frac{y-2}{1-3} = -\frac{1}{5}$$

$$\frac{y-2}{-2} = -\frac{1}{5}$$

$$y - 2 = \frac{2}{5}$$

$$y = 2\frac{2}{5}$$

The student writes the mixed number as the decimal, showing point B is located at (1, 2.4).

## Sample Response: 1 point

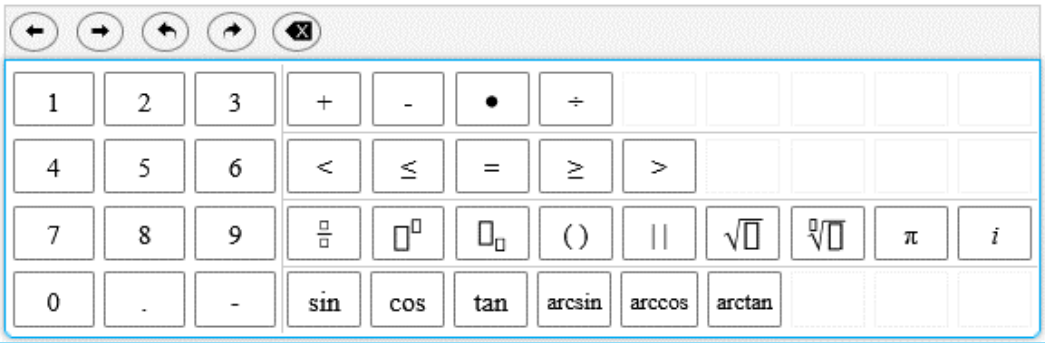
Triangle ABC, with the following characteristics, is placed on the coordinate plane.

- $\overline{AB}$ is on a vertical line.
- $\angle C$ is a right angle.
- Point C is located at (3, 2).
- The slope of $\overline{AC}$ is 5.

What are possible coordinates for points A and B?

A ( 1 , −8 )

B ( −7 , 4 )



**Notes on Scoring**

This response earns partial credit (1 point) because while the student finds points A and B such that AC is perpendicular to BC, A and B are not on the same vertical line.

The student finds the coordinates for point A by graphing point C at (3, 2) and then using the given information that the slope of $\overline{\mathrm{AC}}$ is 5. Since slope is defined as the change in *y* divided by the change in x, a slope of 5 represents the change of 5 in the *y*-direction and a change of 1 in the x-direction. This pair of changes is applied twice from point C, moving down 10 units and left 2 units to identify a possible location for point A at (1, –8).

The student disregards that Points A and B are on a vertical and instead finds the coordinates for point B just using the fact that angle C is a right angle. This means that $\overline{\mathrm{AC}}$ and $\overline{\mathrm{BC}}$ are perpendicular and that the product of their slopes is $-1$. Since the slope of $\overline{\mathrm{AC}}$ is 5, the slope of $\overline{\mathrm{BC}}$ must be $-\frac{1}{5}$ so that the product of these two slopes is $-1$. The slope of $\overline{\mathrm{BC}}$ is then applied twice from point C by moving up 2 units and left 10 units to identify a possible location for point B at (–7, 4).

## Sample Response: 1 point

Triangle ABC, with the following characteristics, is placed on the coordinate plane.

- $\overline{AB}$ is on a vertical line.
- $\angle C$ is a right angle.
- Point C is located at (3, 2).
- The slope of $\overline{AC}$ is 5.

What are possible coordinates for points A and B?

A ( 4 , 7 )

B ( 4 , $2\frac{1}{5}$ )



**Notes on Scoring**

This response earns partial credit (1 point) because the student correctly identifies that the x-coordinates for points A and B must be the same to create a vertical line but does not apply the other given information.

The student finds the coordinates for point A by graphing point C at (3, 2) and then using the given information that the slope of $\overline{\mathrm{AC}}$ is 5. Since a slope of 5 represents the change of 5 in the *y*-direction and a change of 1 in the x-direction point A is located by moving up 5 units and right 1 unit to identify a possible location for point A at (4, 7).

Since $\overline{\mathrm{AB}}$ is on a vertical line, the student understands that points A and B have the same x-coordinate, 4. To find the *y*-coordinate for point B, the student recalls the given information that angle C is a right angle, or that $\overline{\mathrm{AC}}$ and $\overline{\mathrm{BC}}$ are perpendicular, which means the product of their slopes is $-1$. Since the slope of $\overline{\mathrm{AC}}$ is 5, the slope of $\overline{\mathrm{BC}}$ must be $-\frac{1}{5}$. The student may incorrectly solve the slope formula when calculating the *y*-coordinate of point B by adding 2 to both sides when trying to isolate the *y* in the equation $2 - y = \frac{1}{5}$.

$$\frac{2-y}{3-4} = -\frac{1}{5}$$

$$\frac{2-y}{-1} = -\frac{1}{5}$$

$$2 - y = \frac{1}{5}$$

$$y = 2\frac{1}{5}$$

Showing point B is located at $\left(4, 2\frac{1}{5}\right)$ when it should have been $\left(4, 1\frac{4}{5}\right)$.

## Sample Response: 0 points

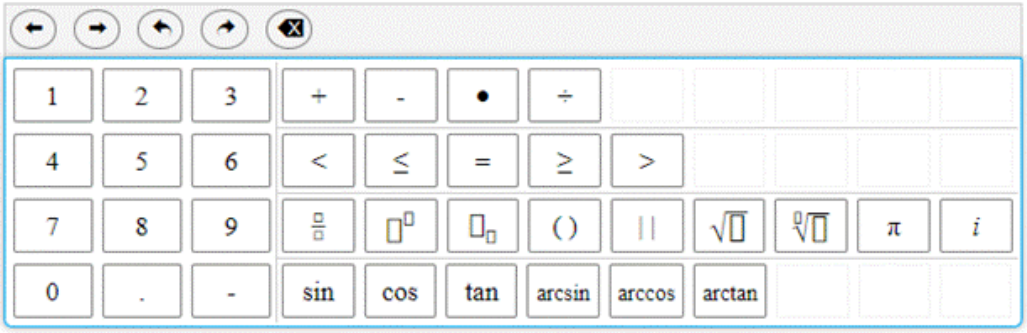
Triangle ABC, with the following characteristics, is placed on the coordinate plane.

- $\overline{AB}$ is on a vertical line.
- $\angle C$ is a right angle.
- Point C is located at (3, 2).
- The slope of $\overline{AC}$ is 5.

What are possible coordinates for points A and B?

A ( 4 , 7 )

B ( 2 , −3 )




**Notes on Scoring**

This response earns no credit (0 points) because the student incorrectly finds the coordinates for points A and B.

The student may only use the information that point C is located at (3, 2) and that the slope of AC is 5. The student may find point A by going up 5 and to the right 1 from point C, resulting in the coordinates (4, 7). The student may then think that the slope of BC should also be 5, and find point B by going down 5 and to the left 1 from point C, resulting in the coordinates (2, –3).

## Sample Response: 0 points

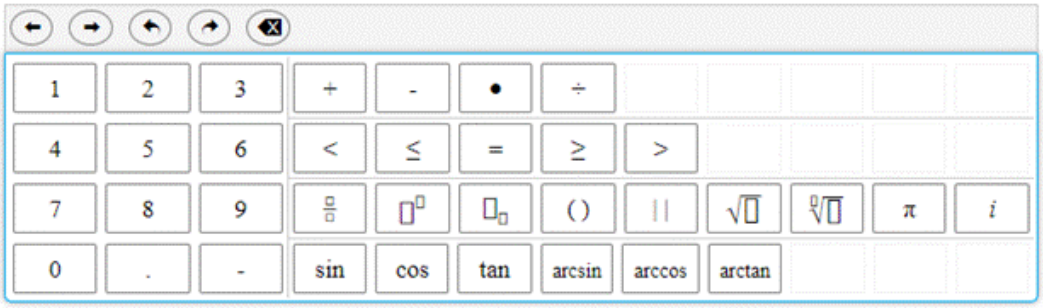
Triangle ABC, with the following characteristics, is placed on the coordinate plane.

- $\overline{AB}$ is on a vertical line.
- $\angle C$ is a right angle.
- Point C is located at (3, 2).
- The slope of $\overline{AC}$ is 5.

What are possible coordinates for points A and B?

A ( 8 , 3 )

B ( −2 , 3 )



**Notes on Scoring**

This response earns no credit (0 points) because the student incorrectly finds the coordinates for points A and B.

The student may think that the slope of 5 represents a change in the *x*-coordinate of 5 and a change in the *y*-coordinate of 1, calculating that point A is at (8, 3). The student may also think that since $\overline{AB}$ is on a vertical line, the *y*-coordinates would be the same because the *y*-axis is a vertical line. The student then solves the slope formula to calculate the x coordinate of point B.

$$\frac{3-2}{x-3} = -\frac{1}{5}$$

$$\frac{1}{x-3} = \frac{1}{-5}$$

$$x-3 = -5$$

$$x = -2$$

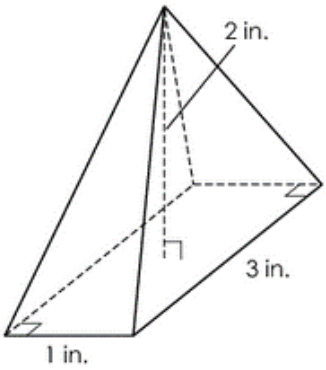
# Geometry
# Spring 2023 Item Release

# Question 7

# Question and Scoring Guidelines

# Question 7

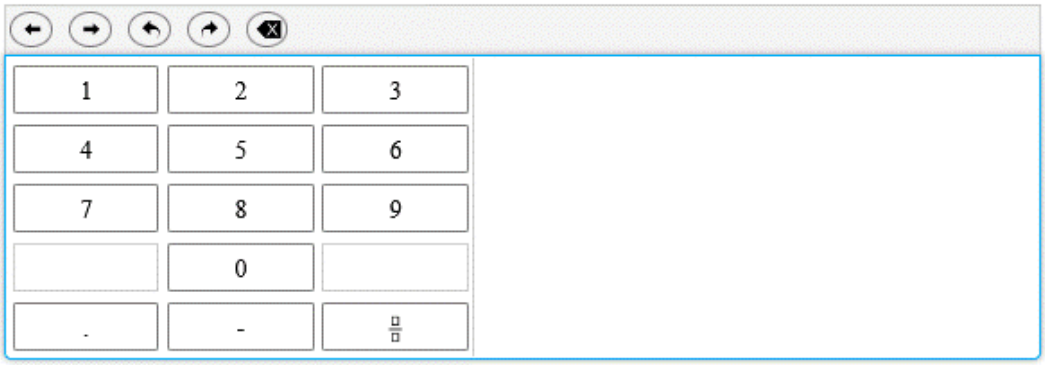
Iris created a pyramid-shaped figurine out of solid aluminum with the dimensions shown.




Iris sells all of her figurines for $7.50 per cubic inch.

What is the price of this figurine?

$



**Points Possible:** 1

**Content Cluster:** Explain volume formulas and use them to solve problems.

**Content Standard:** Use volume formulas for cylinders, pyramids, cones, and spheres to solve problems. *(G.GMD.3)*

**Depth of Knowledge:** Level 2

**Modeling and Reasoning:** Yes

# Scoring Guidelines

For full credit (1 point), the student's response satisfies the bullet below.

- The student calculates the correct price, providing evidence of the ability to use volume formulas for pyramids to solve problems.

Exemplar Response

- 15
  or
- any equivalent value

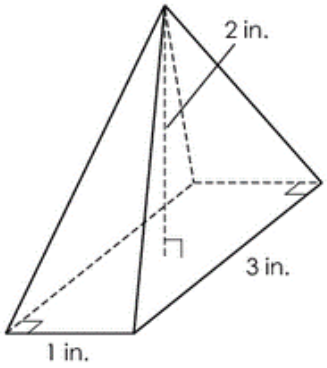
# Geometry
# Spring 2023 Item Release

# Question 7

# Sample Responses

## Sample Response: 1 point

Iris created a pyramid-shaped figurine out of solid aluminum with the dimensions shown.




Iris sells all of her figurines for $7.50 per cubic inch.

What is the price of this figurine?

$ 15



**Notes on Scoring**

This response earns full credit (1 point) because the student correctly finds the volume of the figurine and multiplies the volume by the price per cubic inch.

The student recognizes that the unit price of Iris' figurines is per unit of volume. They use the formula for the volume of a pyramid to calculate the figurine's volume.

$$V = \frac{1}{3} \cdot l \cdot w \cdot h$$

$$= \frac{1}{3} \cdot 1 \cdot 2 \cdot 3$$

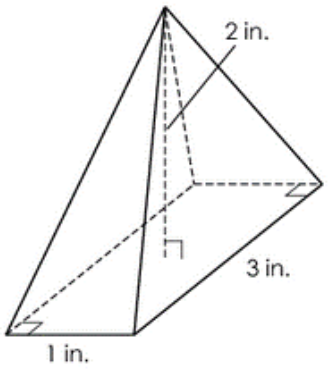
$$= \frac{1}{3} \cdot 6$$

$$= 2$$

To calculate the price of a figurine, the total volume of the figurine, 2 cubic inches, is multiplied by the unit price, \$7.50 per cubic inch, $2 \cdot 7.50 = 15$. Thus, the price of the figurine is \$15.

## Sample Response: 0 points

Iris created a pyramid-shaped figurine out of solid aluminum with the dimensions shown.




Iris sells all of her figurines for $7.50 per cubic inch.

What is the price of this figurine?

$ 2

| | | |
|---|---|---|
| 1 | 2 | 3 |
| 4 | 5 | 6 |
| 7 | 8 | 9 |
| | 0 | |
| . | - | |

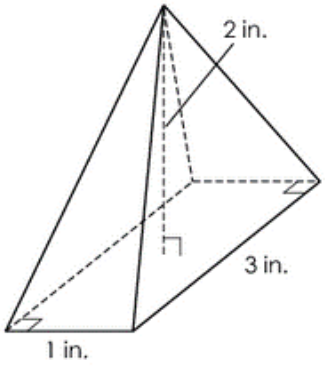
**Notes on Scoring**

This response earns no credit (0 points) because the student may find the volume of the figurine but does not correctly use the volume to find the price of the figurine.

The student may correctly calculate the volume of the figurine and use this as their answer instead of using the volume to calculate the figurine's price.

## Sample Response: 0 points

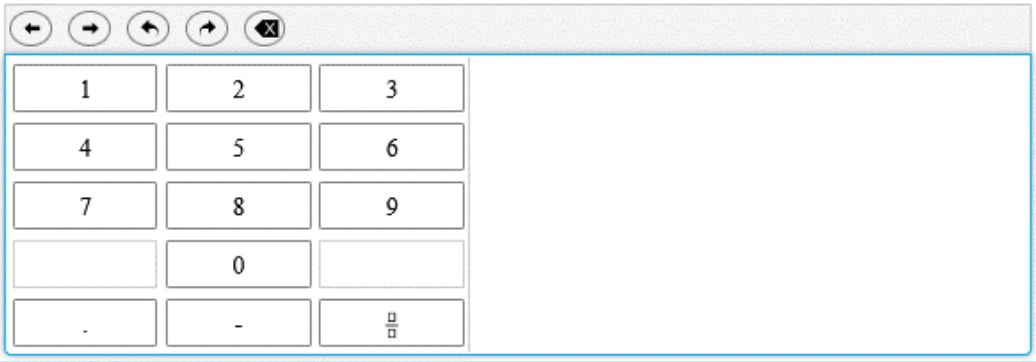
Iris created a pyramid-shaped figurine out of solid aluminum with the dimensions shown.




Iris sells all of her figurines for $7.50 per cubic inch.

What is the price of this figurine?

$ 45



**Notes on Scoring**

This response earns no credit (0 points) because the student finds the incorrect price of the figurine.

The student may incorrectly calculate the volume of the figurine, using the volume formula for a rectangular prism instead of the volume formula for a pyramid.

$$V = l \cdot w \cdot h$$

$$= 1 \cdot 2 \cdot 3$$

$$= 6$$

To find the price of the figurine, the volume is then multiplied by the unit price of the figurine, \$7.50 per cubic inch, $6 \cdot 7.50 = 45$. Thus, the price of the figurine is calculated to be \$45.

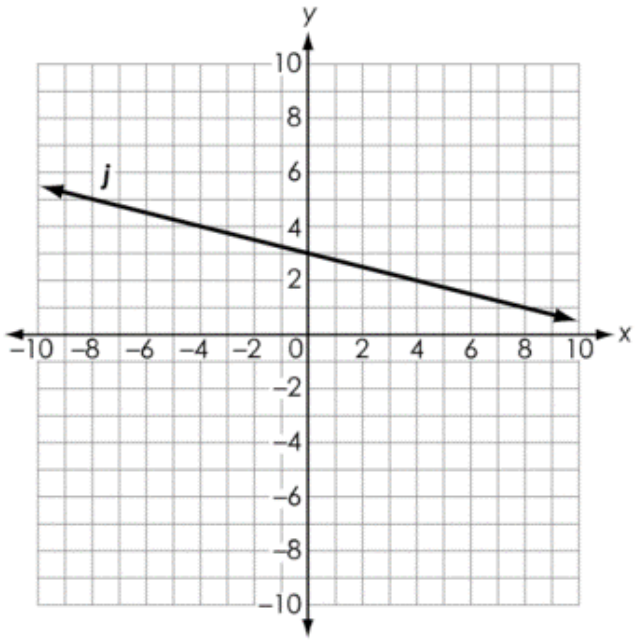
# Geometry
# Spring 2023 Item Release

# Question 9

# Question and Scoring Guidelines

# Question 9

Nasir graphs line $j$ on the coordinate plane as shown.




He dilates $j$ by a scale factor of 3 about point P to create line $k$. Line $j$ and line $k$ coincide.

What is a possible ordered pair for point P?

( , )



**Points Possible:** 1

**Content Cluster:** Understand similarity in terms of similarity transformations.

**Content Standard:** Verify experimentally the properties of dilations given by a center and a scale factor:
*a*. A dilation takes a line not passing through the center of the dilation to a parallel line and leaves a line passing through the center unchanged. *(G.SRT.1)*

**Depth of Knowledge:** Level 2

**Modeling and Reasoning:** Yes

# Scoring Guidelines

For full credit (1 point), the student's response satisfies the bullet below.

- The student correctly identifies a possible center of dilation, providing evidence of the understanding that a dilation takes a line not passing through the center of the dilation to a parallel line, and leaves a line passing through the center unchanged.

Exemplar Response

- (0, 3)
  or
- any coordinate pairs satisfying the equation of line *j*, $y = {}^{-1}/_{(4)}\,x + 3$

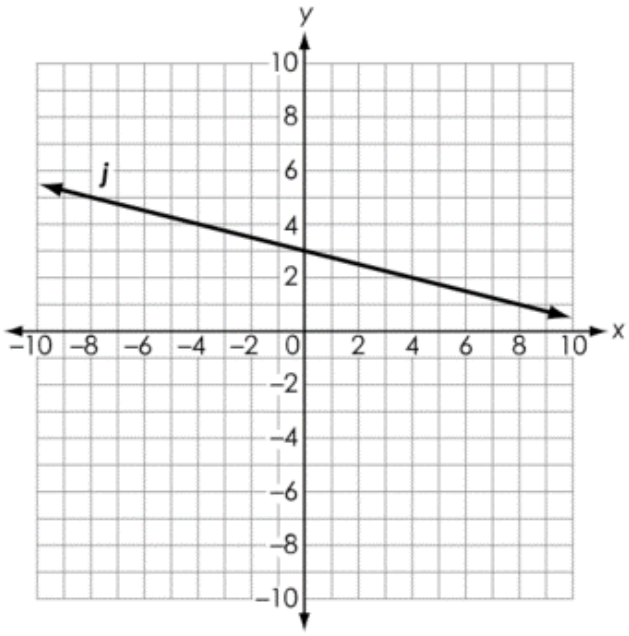
# Geometry
# Spring 2023 Item Release

## Question 9

## Sample Responses

## Sample Response: 1 point

Nasir graphs line $j$ on the coordinate plane as shown.




He dilates $j$ by a scale factor of 3 about point P to create line $k$. Line $j$ and line $k$ coincide.

What is a possible ordered pair for point P?

( 0 , 3 )



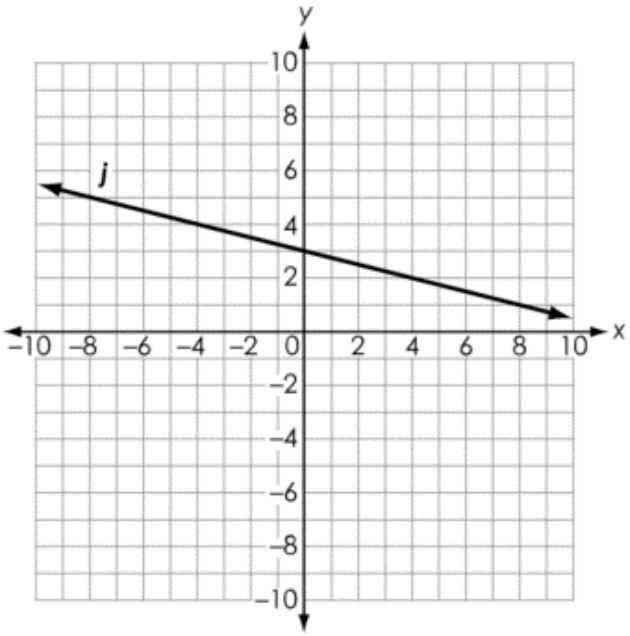
**Notes on Scoring**

This response earns full credit (1 point) because the student correctly finds the ordered pair for point P at (0, 3) so that lines *j* and *k* coincide after dilating line *j* by a scale factor of 3.

The student recognizes since line *j* and line *k* coincide, the two lines must lie on top of one another. To show a dilation of line *k* directly on top of line *j*, a possible ordered pair for the center of dilation, point P, is any point located on line *j*. The student identifies the point (0, 3) as being on line *j* and thus a possible location for point P.

## Sample Response: 1 point

Nasir graphs line $j$ on the coordinate plane as shown.




He dilates $j$ by a scale factor of 3 about point P to create line $k$. Line $j$ and line $k$ coincide.

What is a possible ordered pair for point P?

( 4 , 2 )



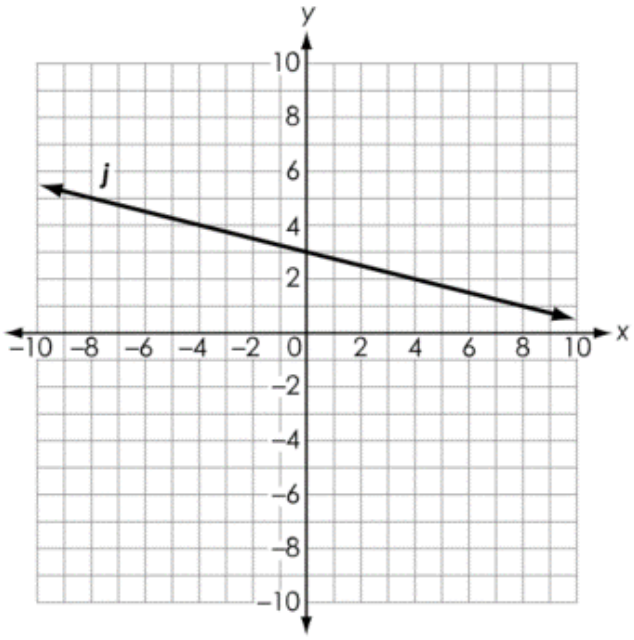
**Notes on Scoring**

This response earns full credit (1 point) because the student correctly finds the ordered pair for point P at (4, 2) so that lines *j* and *k* coincide after dilating line *j* by a scale factor of 3.

The student recognizes since line *j* and line *k* coincide, the two lines must lie on top of one another. To show a dilation of line *k* directly on top of line *j*, a possible ordered pair for the center of dilation, point P, is any point located on line *j*. The student identifies the point (4, 2) as being on line *j* and thus a possible location for point P.

## Sample Response: 0 points

Nasir graphs line $j$ on the coordinate plane as shown.




He dilates $j$ by a scale factor of 3 about point P to create line $k$. Line $j$ and line $k$ coincide.

What is a possible ordered pair for point P?

( 0 , 0 )

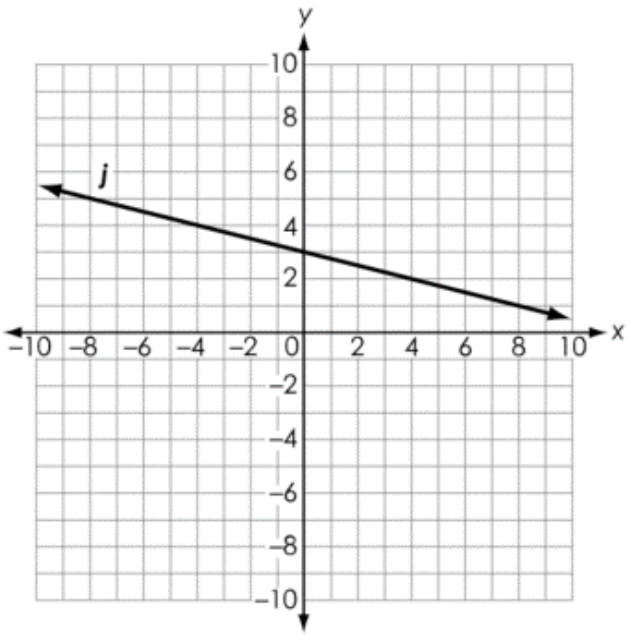
**Notes on Scoring**

This response earns no credit (0 points) because the student incorrectly finds the ordered pair for point P at (0, 0), not creating coinciding lines *j* and *k* after dilating line *j* by a scale factor of 3.

The student may think to show lines *j* and *k* coincide, the two lines must be parallel, not understanding the definition of coincide. To show a dilation of line *k* so that line *j* is parallel, a possible ordered pair for point P is located at the origin (0, 0) and any point not on line *j* as the center of dilation will result in parallel lines.

## Sample Response: 0 points

Nasir graphs line $j$ on the coordinate plane as shown.




He dilates $j$ by a scale factor of 3 about point P to create line $k$. Line $j$ and line $k$ coincide.

What is a possible ordered pair for point P?

( 10 , 0 )

**Notes on Scoring**

This response earns no credit (0 points) because the student incorrectly finds the ordered pair for point P at (10, 0), not creating coinciding lines *j* and *k* after dilating line *j* by a scale factor of 3.

The student may think to show lines *j* and *k* coincide, the two lines must be parallel, not understanding the definition of coincide. To show a dilation of line *k* so that line *j* is parallel, a possible ordered pair for point P is located at (10, 0) and any point not on line *j* as the center of dilation will result in parallel lines.

# Geometry
# Spring 2023 Item Release

# Question 11

# Question and Scoring Guidelines

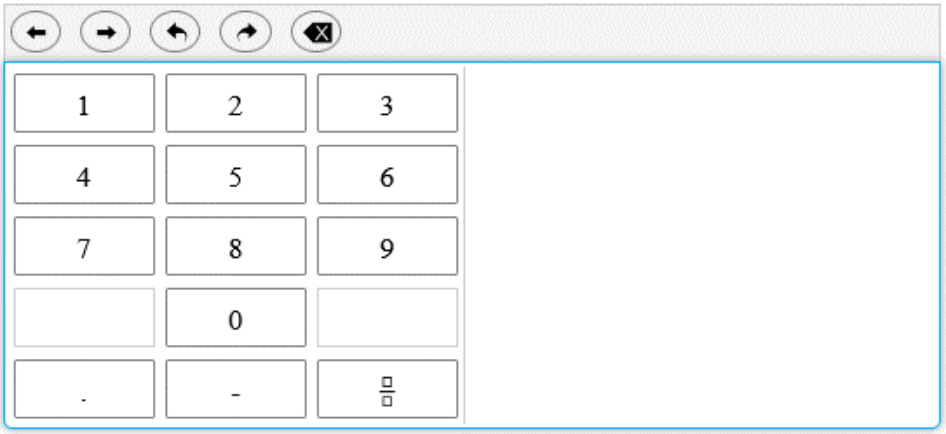
# Question 11

Olivia creates a logo made up of three triangles. She constructs $\triangle ABC$ with an area of 6 square units.

- She reflects $\triangle ABC$ across $\overline{BC}$ to form $\triangle DBC$.
- She then reflects $\triangle DBC$ across $\overline{DC}$ to form $\triangle DEC$.

What is the total area, in square units, of Olivia's logo?

square units




**Points Possible:** 1

**Content Cluster:** Understand congruence in terms of rigid motions.

**Content Standard:** Use geometric descriptions of rigid motions to transform figures and to predict the effect of a given rigid motion on a given figure; given two figures, use the definition of congruence in terms of rigid motions to decide if they are congruent. *(G.CO.6)*

**Depth of Knowledge:** Level 2

**Modeling and Reasoning:** No

# Scoring Guidelines

For full credit (1 point), the student's response satisfies the bullet below.

- The student calculates the correct area of the figure, providing evidence of the ability to use geometric descriptions of rigid motions to transform figures and to predict the effect of a given set of rigid motions on a given figure.

Exemplar Response

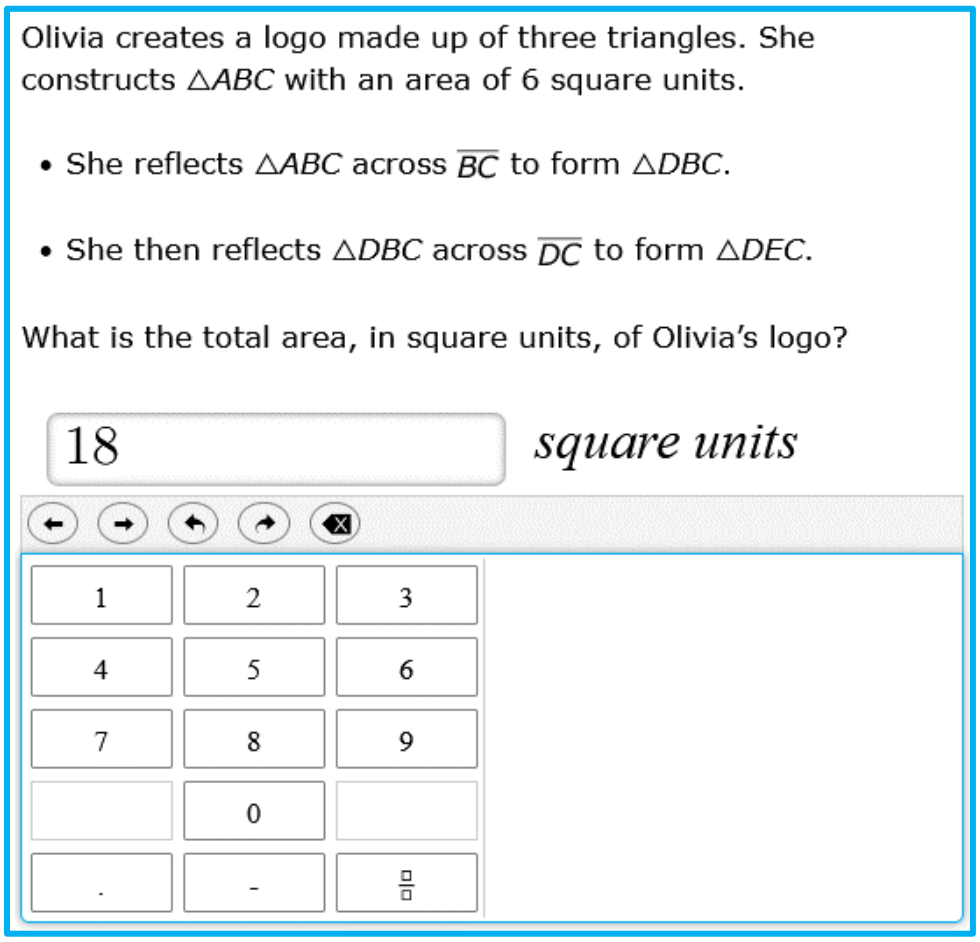
- 18 square units
  or
- any equivalent value

# Geometry
# Spring 2023 Item Release

# Question 11

# Sample Responses

## Sample Response: 1 point

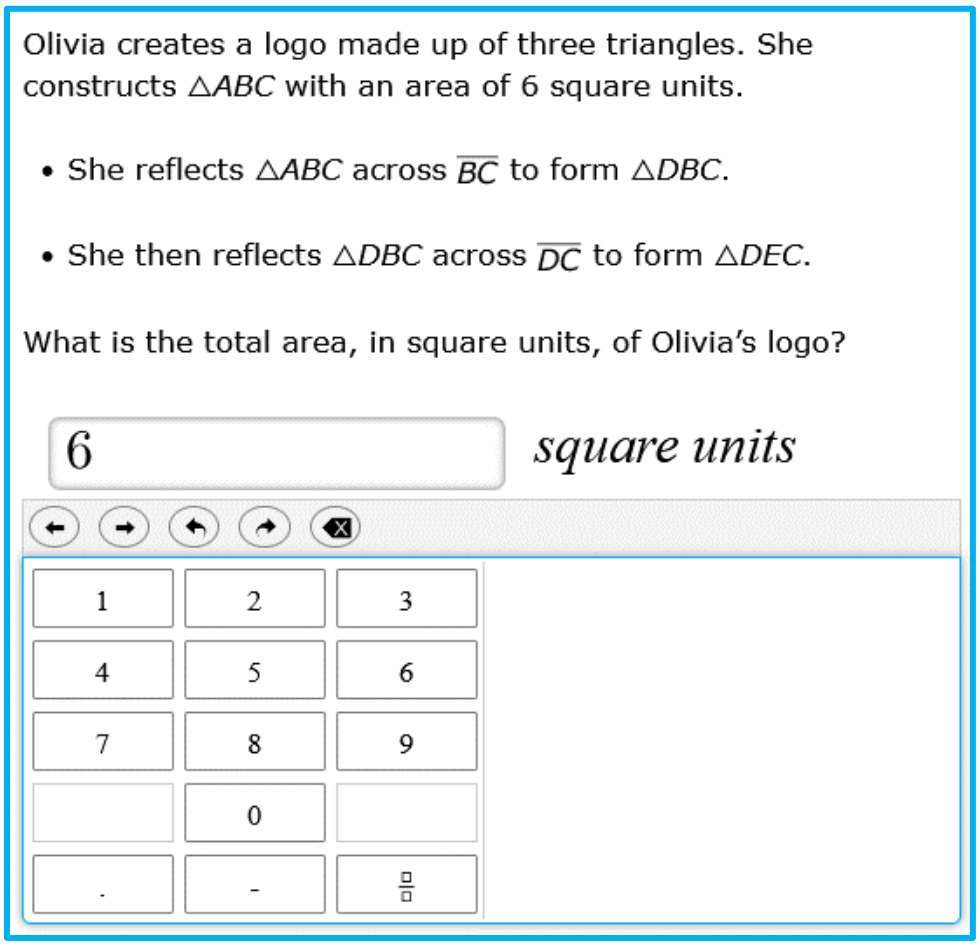



**Notes on Scoring**

This response earns full credit (1 point) because the student correctly finds the total area of all three triangles that create Olivia's logo.

The student understands to find Olivia's logo, they must find the area of each of the 3 triangles. Since reflections preserve angle measure and length, the student may conclude all three triangles are congruent and therefore have the same area. Since the area of triangle ABC is 6 square units, triangle DBC has an area of 6 square units, and triangle DEC also has an area of 6 square units. The area of each individual triangle is added together to calculate the area of Olivia's logo as 18 square units.

## Sample Response: 0 points

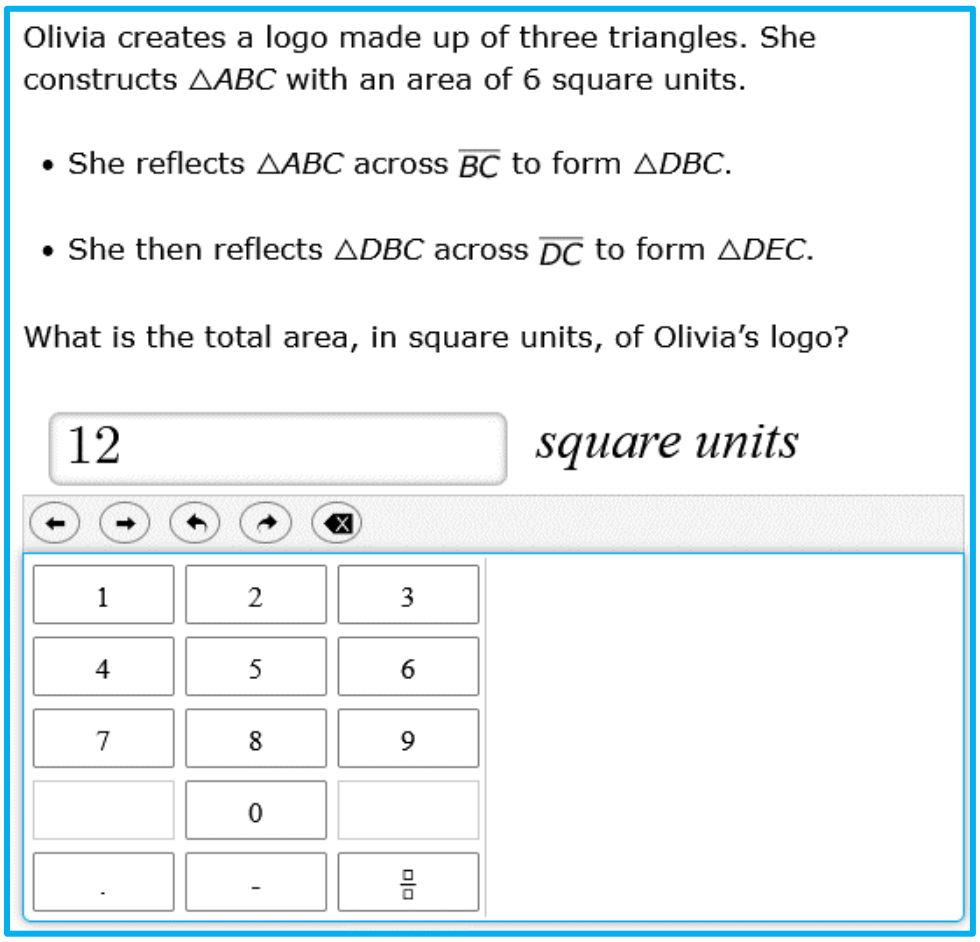



**Notes on Scoring**

This response earns no credit (0 points) because the student incorrectly finds the total area of Olivia's logo.

The student may not understand that the area of the logo is the sum of the three triangles and gives the area of each triangle.

## Sample Response: 0 points




**Notes on Scoring**

This response earns no credit (0 points) because the student finds the area for two triangles, calculating only part of the area of Olivia's logo.

The student may misinterpret the area of Olivia's logo as 2 triangles, triangle DBC and triangle DEC. They correctly recall that since reflections preserve angle measure and length, triangle ABC, triangle DBC, and triangle DEC are congruent and have the same area, so the area of triangle DBC and triangle DEC is 6 square units. The student adds the areas of the two transformed triangles to find the area of Olivia's logo as 12 square units and overlooks the fact that the logo includes triangle ABC as well.

# Geometry
# Spring 2023 Item Release

# Question 16

# Question and Scoring Guidelines

## Question 16

Which statement most accurately defines a ray?

(A) all points of a line that are on either side of a fixed point

(B) all points of a line that are on one side of a fixed point including the fixed point

(C) all points of a line that are between two fixed points of the line

(D) all points of a line that are between two fixed points of the line including the two fixed points

**Points Possible:** 1

**Content Cluster:** Experiment with transformations in the plane.

**Content Standard:** Know precise definitions of ray, angle, circle, perpendicular line, parallel line, and line segment, based on the undefined notions of point, line, distance along a line, and arc length. *(G.CO.1)*

**Depth of Knowledge:** Level 1

**Modeling and Reasoning:** No

# Scoring Guidelines

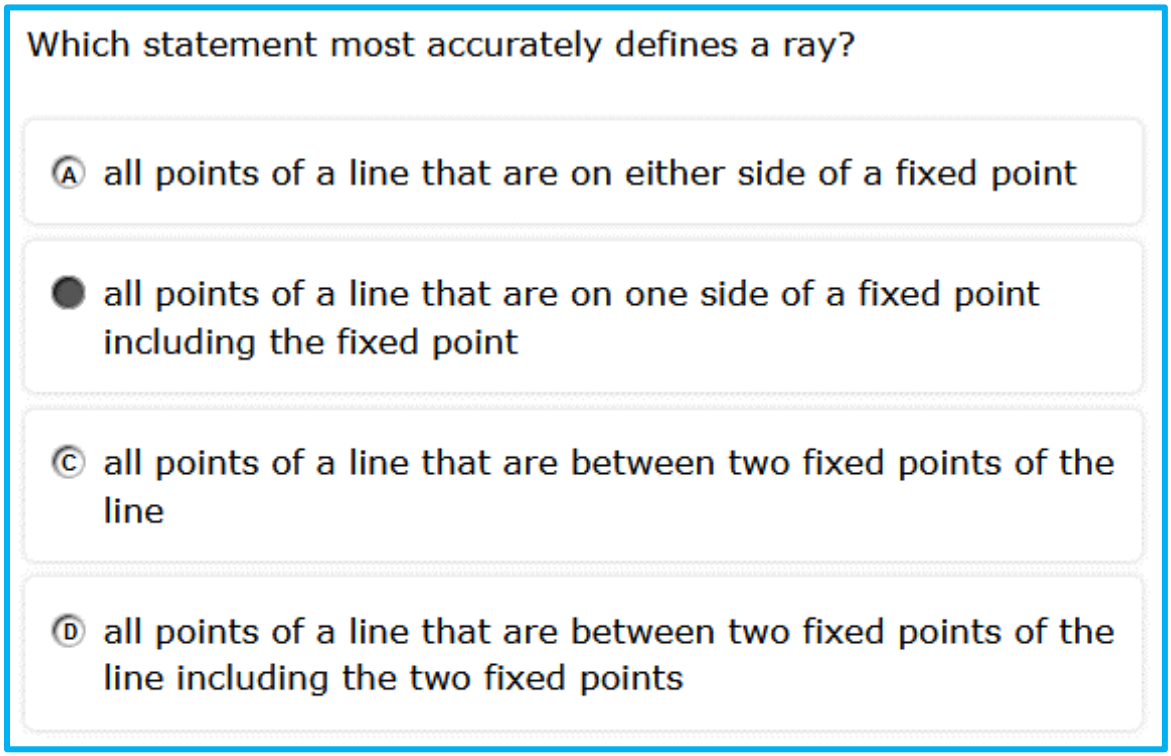
Rationale for Option A: This is incorrect. The student may realize that a ray contains a fixed point but may not understand that if all points on both sides of the fixed point are included, then the resulting figure is an entire line instead of a ray.

Rationale for Option B: **Key** – The student identifies the definition of a ray.

Rationale for Option C: This is incorrect. The student may confuse the definition of a ray with an incomplete definition of a line segment that should include the two fixed points (endpoints of a line segment).

Rationale for Option D: This is incorrect. The student may confuse the definition of a ray with the definition of a line segment.

## Sample Response: 1 point



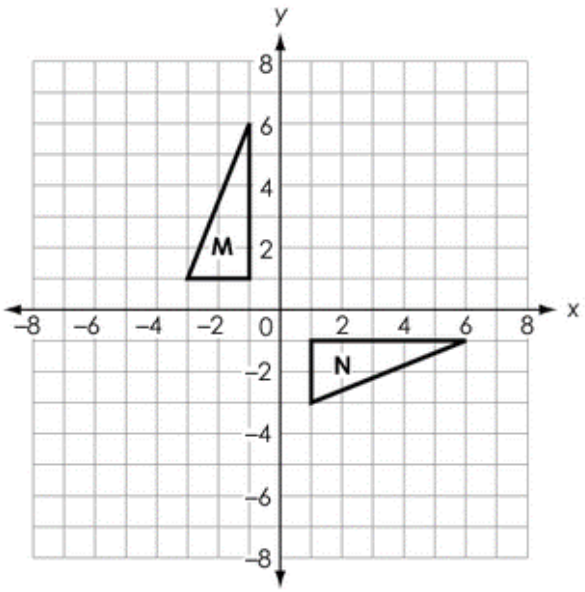
# Geometry
# Spring 2023 Item Release

# Question 18

# Question and Scoring Guidelines

# Question 18

A transformation is performed on triangle M to create triangle N, as shown.




Which transformation was performed?

Ⓐ a reflection across the line $y = -x$

Ⓑ a reflection across the line $y = x$

Ⓒ a clockwise rotation of 180° about the origin

Ⓓ a clockwise rotation of 270° about the origin

**Points Possible:** 1

**Content Cluster:** Experiment with transformations in the plane.

**Content Standard:** Given a geometric figure and a rotation, reflection, or translation, draw the transformed figure using items such as graph paper, tracing paper, or geometry software. Specify a sequence of transformations that will carry a given figure onto another. *(G.CO.5)*

**Depth of Knowledge:** Level 2

**Modeling and Reasoning:** Yes

# Scoring Guidelines

Rationale for Option A: This is incorrect. The student may confuse the line $y = x$ with $y = –x$, and think the values for x- and y-coordinates switch showing a reflection across $y = –x$.

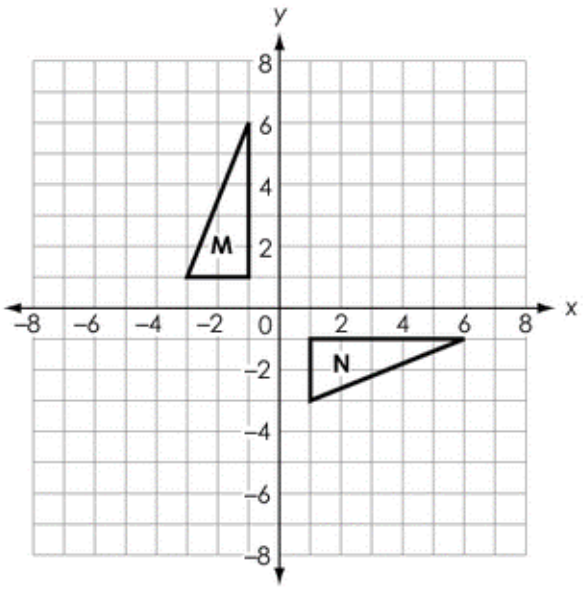
Rationale for Option B: **Key** – The student notices that the x- and y-coordinates switch, meaning it reflects across $y = x$.

Rationale for Option C: This is incorrect. The student may notice that a 180-degree clockwise rotation would move the triangle to Quadrant IV but forget that the triangle would be "pointing" downward, not to the right, if it represents a rotation.

Rationale for Option D: This is incorrect. The student may notice that a 270-degree counterclockwise rotation moves shapes in Quadrant I into Quadrant IV, but does not realize that triangle M is in Quadrant II and the rotation is clockwise. The result of the rotation on triangle M would be a triangle in Quadrant III.

## Sample Response: 1 point

A transformation is performed on triangle M to create triangle N, as shown.




Which transformation was performed?

Ⓐ a reflection across the line $y = -x$

● a reflection across the line $y = x$

Ⓒ a clockwise rotation of 180° about the origin

Ⓓ a clockwise rotation of 270° about the origin

# Geometry
# Spring 2023 Item Release

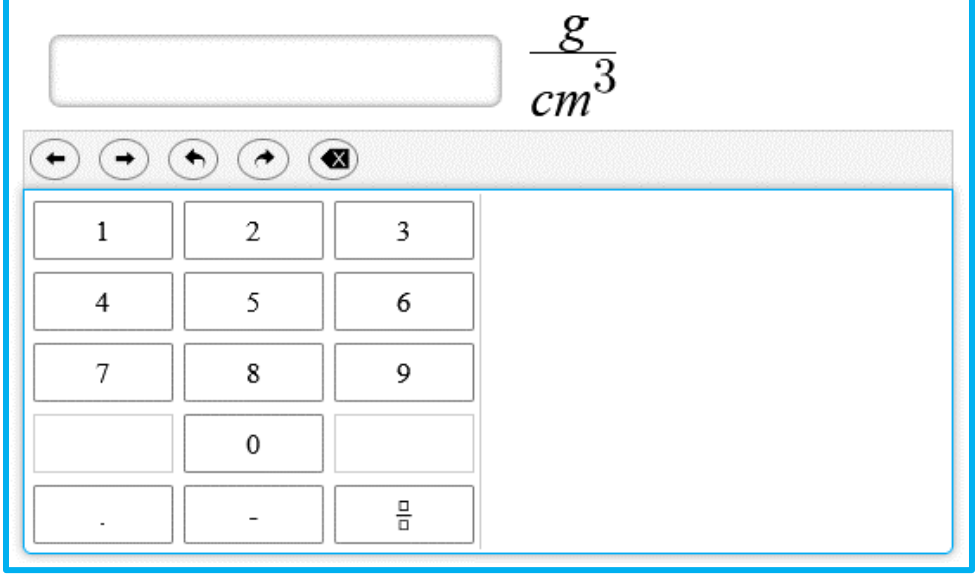
# Question 27

# Question and Scoring Guidelines

# Question 27

A brick has a mass of 2,022.75 grams and a volume of 1,064.5 cubic centimeters.

What is the density of the brick, in grams per cubic centimeter $\left(\frac{g}{cm^3}\right)$? Round your answer to the nearest tenth.




**Points Possible:** 1

**Content Cluster:** Apply geometric concepts in modeling situations.

**Content Standard:** Apply concepts of density based on area and volume in modeling situations, e.g., persons per square mile, BTUs per cubic foot.★ *(G.MG.2)*

**Depth of Knowledge:** Level 1

**Modeling and Reasoning:** Yes

# Scoring Guidelines

For full credit (1 point), the student's response satisfies the bullet below.

- The student calculates the correct density, providing evidence of the ability to apply concepts of density based on area and volume in modeling situations.

Exemplar Response

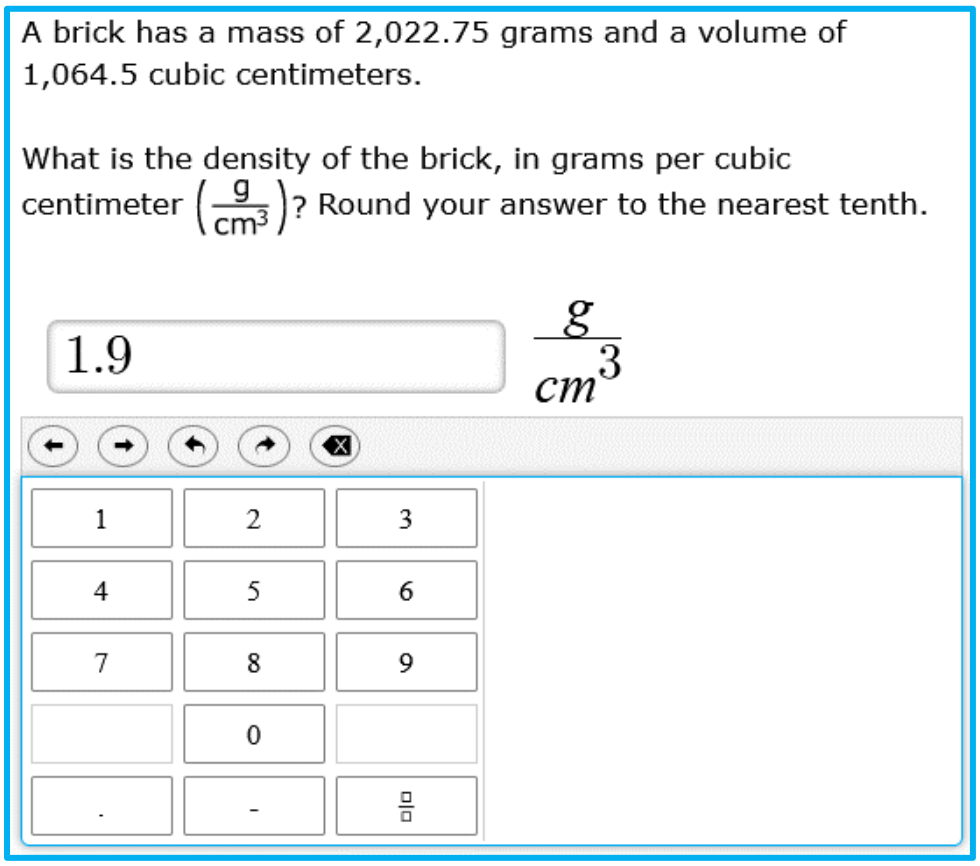
- 1.9 $\frac{g}{cm^3}$
  or
- any value in the range 1.9 to 1.9002

# Geometry
# Spring 2023 Item Release

# Question 27

# Sample Responses

## Sample Response: 1 point




**Notes on Scoring**

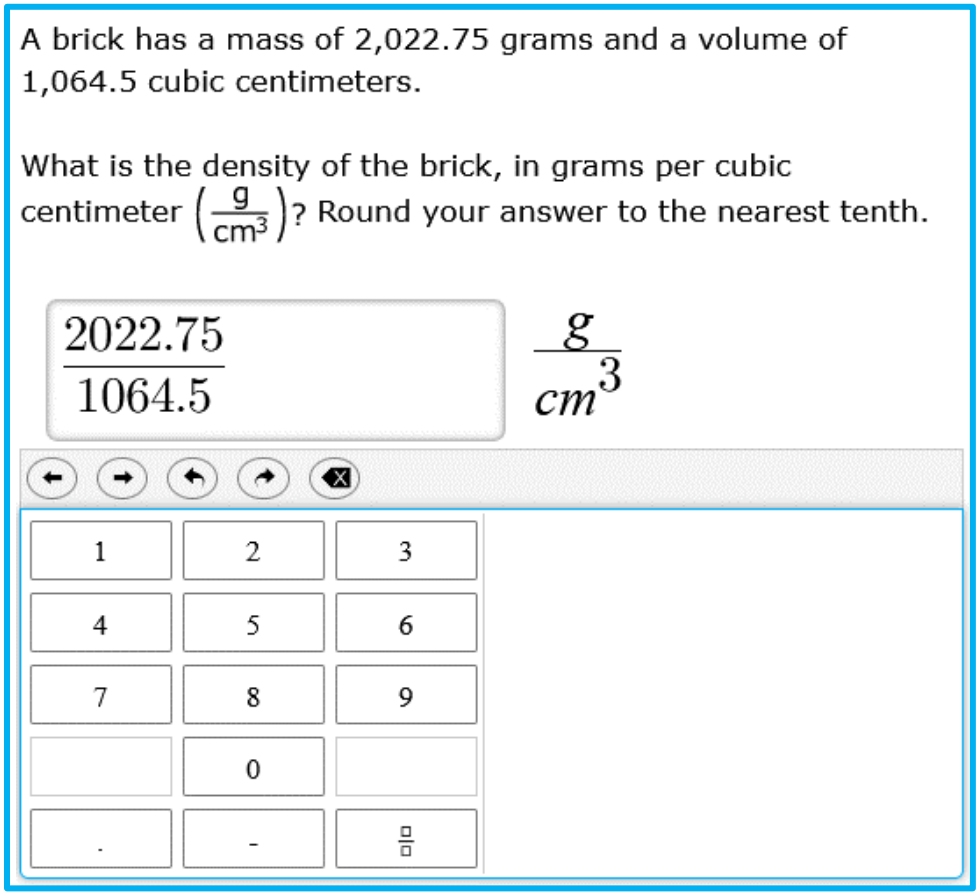
This response earns full credit (1 point) because the student correctly finds the density of the brick in grams per cubic centimeter.

The student understands since the density of the brick is measured in grams per cubic centimeter, so the mass of the brick is divided by the volume of the brick.

$$\frac{2{,}022.75}{1{,}064.5} \approx 1.9002$$

$$\approx 1.9$$

## Sample Response: 1 point

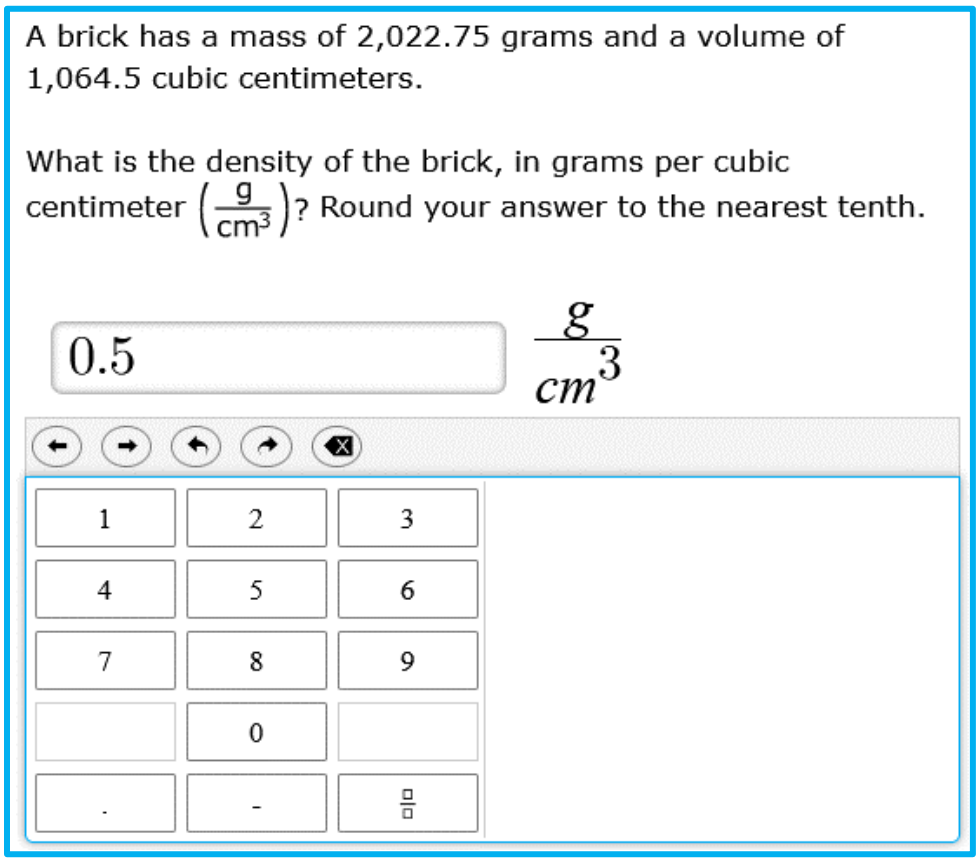



**Notes on Scoring**

This response earns full credit (1 point) because the student correctly finds the density of the brick in grams per cubic centimeter.

The student understands since the density of the brick is measured in grams per cubic centimeter, so the mass of the brick is divided by the volume of the brick. The student identifies the mass and volume of the brick and uses those two numbers to write the fraction $\frac{2,022.75}{1,064.5}$, not performing the division.

## Sample Response: 0 points




**Notes on Scoring**

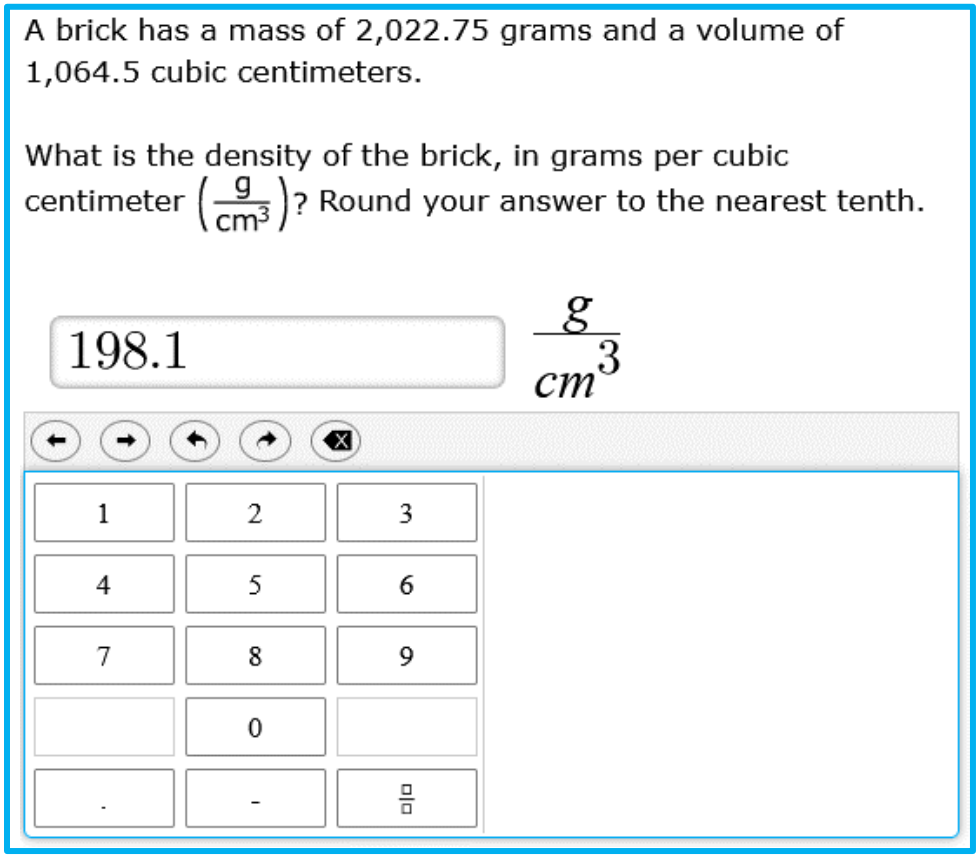
This response earns no credit (0 points) because the student incorrectly calculates the density of the brick in grams per cubic centimeter.

The student may understand the density of the brick is measured in grams per cubic centimeter but thinks the volume of the brick should be divided by the mass of the brick.

$$\frac{1{,}064.5}{2{,}022.75} \approx 0.5263$$

$$\approx 0.5$$

## Sample Response: 0 points




**Notes on Scoring**

This response earns no credit (0 points) because the student incorrectly calculates the density of the brick in grams per cubic centimeter.

The student may think to find the density of the brick, they must first take the cubic root of the volume since its units are cubic centimeters, $\sqrt[3]{1{,}064.5} \approx 10.21$. The student then thinks they can find the density of the brick by taking the mass of the brick and dividing it by 10.21 centimeters.

$$\frac{2{,}022.75}{10.21} \approx 98.1146$$

$$\approx 198.1$$

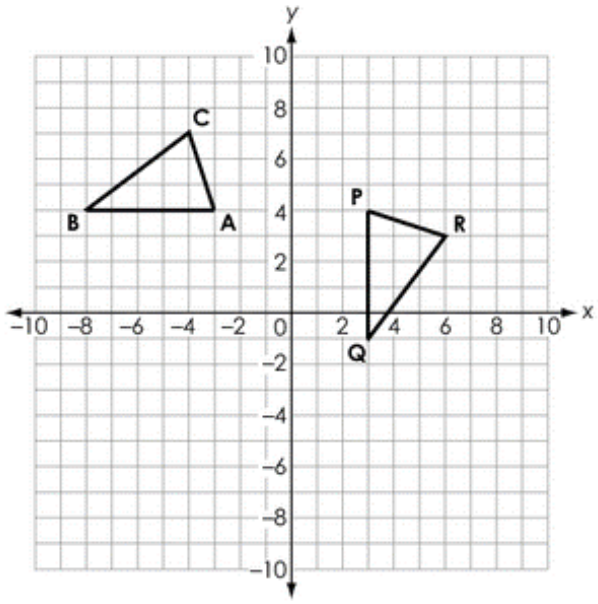
# Geometry
# Spring 2023 Item Release

# Question 28

# Question and Scoring Guidelines

# Question 28

Triangles ABC and PQR are shown on the coordinate grid.




Robby wants to show that the two triangles are congruent.

Which sequence of transformations can Robby apply to triangle ABC to show that it is congruent to triangle PQR?

Ⓐ a reflection of triangle ABC across the *y*-axis, then a translation of the image 5 units down

Ⓑ a rotation of triangle ABC 90° clockwise about point A, then a reflection of the image across the *x*-axis

Ⓒ a rotation of triangle ABC 90° clockwise about point A, then a translation of the image 6 units to the right

Ⓓ a reflection of triangle ABC across the *y*-axis, then a rotation of the image 90° clockwise about the image of point A

**Points Possible:** 1

**Content Cluster:** Understand congruence in terms of rigid motions.

**Content Standard:** Use the definition of congruence in terms of rigid motions to show that two triangles are congruent if and only if corresponding pairs of sides and corresponding pairs of angles are congruent. *(G.CO.7)*

**Depth of Knowledge:** Level 2

**Modeling and Reasoning:** No

# Scoring Guidelines

Rationale for Option A: This is incorrect. The student may notice a reflection needs to occur to have the orientation of triangle ABC match the orientation of triangle PQR, and a translation of the image 5 units down shows vertex A coinciding with vertex Q. However, the student does not realize that to prove congruence, all vertices must coincide, and this pair of transformations does not map corresponding vertices.

Rationale for Option B: This is incorrect. The student may notice a rotation and reflection need to occur to have the orientation of triangle ABC match the orientation of triangle PQR. However, the student does not realize that correcting the orientation of triangle ABC does not prove congruence alone. The vertices of the two triangles must coincide to show congruence.

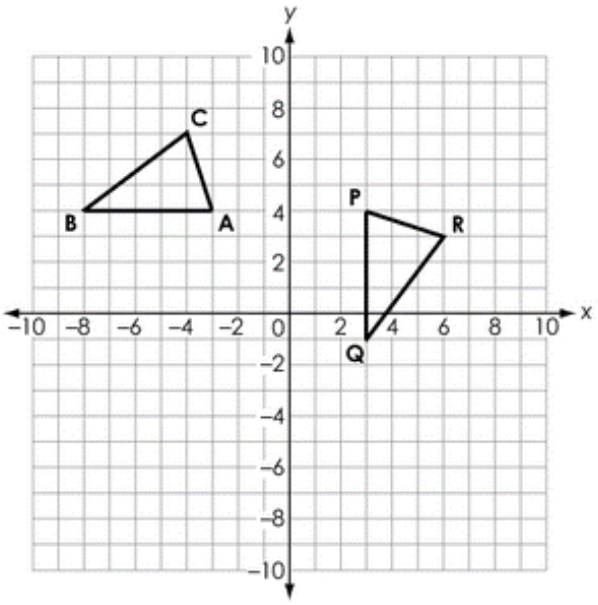
Rationale for Option C: This is incorrect. The student may realize a rotation needs to occur to have the orientation of triangle ABC match the orientation of triangle PQR, and a translation 6 units to the right shows vertex A' coinciding with vertex P. However, the student does not realize that to prove congruence, all vertices must coincide, and this pair of transformations only maps 1 pair of corresponding vertices.

Rationale for Option D: **Key** – The student realizes that a reflection of triangle ABC across the *y*-axis followed by a 90-degree clockwise rotation about point A makes the vertices of the two triangles coincide, thus showing that the two triangles are congruent.

A $(-3, 4)$→ A' $(-(-3), 4) = (3, 4)$ → A'' $(3, 4) = P$
B $(-8, 4)$ → B' $(-(-8),\ 4) = (8,\ 4)$ → B'' $(4 - 1,\ -8 + 7) = (3, -1) = Q$
C $(-4, 7)$ → C' $(-(-4), 7) = (4, 7)$ → C'' $(7 - 1, -4 + 7) = (6, 3) = R$

## Sample Response: 1 point

Triangles ABC and PQR are shown on the coordinate grid.




Robby wants to show that the two triangles are congruent.

Which sequence of transformations can Robby apply to triangle ABC to show that it is congruent to triangle PQR?

Ⓐ a reflection of triangle ABC across the *y*-axis, then a translation of the image 5 units down

Ⓑ a rotation of triangle ABC 90° clockwise about point A, then a reflection of the image across the *x*-axis

Ⓒ a rotation of triangle ABC 90° clockwise about point A, then a translation of the image 6 units to the right

● a reflection of triangle ABC across the *y*-axis, then a rotation of the image 90° clockwise about the image of point A

# Geometry
# Spring 2023 Item Release

# Question 29

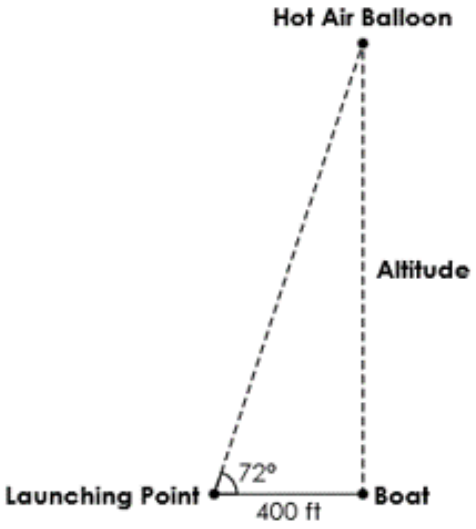
# Question and Scoring Guidelines

# Question 29

A hot air balloon is launched and drifts over a boat.

- The balloon's altitude increases at a constant rate.
- The balloon is directly above a boat 400 feet (ft) away from the launching point 5 minutes after it is launched.
- The angle of elevation is 72°.

A figure describing the situation is shown.




What is the change in the altitude, in feet, of the hot air balloon each minute? Round your answer to the nearest foot.

feet

**Points Possible:** 1

**Content Cluster:** Define trigonometric ratios and solve problems involving right triangles.

**Content Standard:** Solve problems involving right triangles.★
*a.* Use trigonometric ratios and the Pythagorean Theorem to solve right triangles in applied problems if one of the two acute angles and a side length is given. *(G.SRT.8)*

**Depth of Knowledge:** Level 3

**Modeling and Reasoning:** Yes

# Scoring Guidelines

For full credit (1 point), the student's response satisfies the bullet below.

- The student correctly calculates that the altitude of the hot air balloon changes by about 246 feet every minute, providing evidence of the ability to use trigonometric ratios to solve right triangles in applied problems.

Exemplar Response

- 246 feet
  or
- any value between 246 and 246.22, inclusive

# Geometry
# Spring 2023 Item Release

## Question 29

## Sample Responses

## Sample Response: 1 point

A hot air balloon is launched and drifts over a boat.

- The balloon's altitude increases at a constant rate.
- The balloon is directly above a boat 400 feet (ft) away from the launching point 5 minutes after it is launched.
- The angle of elevation is 72°.

A figure describing the situation is shown.




What is the change in the altitude, in feet, of the hot air balloon each minute? Round your answer to the nearest foot.

246 *feet*

**Notes on Scoring**

This response earns full credit (1 point) because the student correctly finds the change in altitude of the hot air balloon per minute using the tangent function.

The student recognizes the distances between the launching point, hot air balloon, and boat form a right triangle, so trigonometry can be used to calculate the altitude. Since the angle of elevation and the adjacent side of the triangle are given, and the opposite side of the triangle is the unknown, the tangent function should be used to calculate the altitude.

$$\begin{aligned} \tan\theta &= \frac{opposite}{adjacent} \\ \tan(72) &= \frac{x}{400} \\ 400\tan(72) &= x \end{aligned}$$

To find the change in altitude of the hot air balloon per minute, the hot air balloon's altitude is divided by the amount of time it has been away from the launching point.

$$\frac{400\tan(72)}{5} \approx 246.214683$$

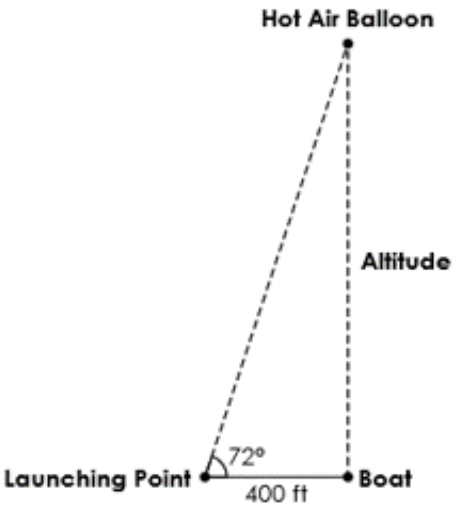
Thus, the change in altitude of the hot air balloon each minute is about 246 feet.

## Sample Response: 1 point

A hot air balloon is launched and drifts over a boat.

- The balloon's altitude increases at a constant rate.
- The balloon is directly above a boat 400 feet (ft) away from the launching point 5 minutes after it is launched.
- The angle of elevation is 72°.

A figure describing the situation is shown.




What is the change in the altitude, in feet, of the hot air balloon each minute? Round your answer to the nearest foot.

246.22 *feet*

**Notes on Scoring**

This response earns full credit (1 point) because the student correctly finds the change in altitude of the hot air balloon per minute using the tangent function.

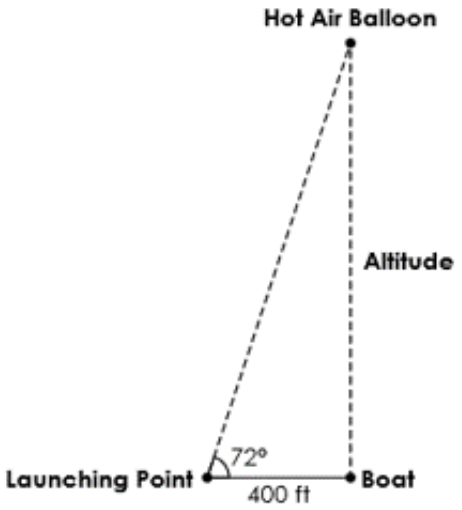
The student recognizes the distances between the launching point, hot air balloon, and boat form a right triangle, so trigonometry can be used to calculate the altitude. Since the angle of elevation and the adjacent side of the triangle are given and the opposite side of the triangle is the unknown, the tangent function should be used to calculate the altitude. The altitude is then divided by 5 to calculate the change in the altitude each minute. The student correctly calculates the rate of change and gives a more precise answer by rounding to the hundredth of a foot instead of to the nearest foot.

## Sample Response: 0 points

A hot air balloon is launched and drifts over a boat.

- The balloon's altitude increases at a constant rate.
- The balloon is directly above a boat 400 feet (ft) away from the launching point 5 minutes after it is launched.
- The angle of elevation is 72°.

A figure describing the situation is shown.




What is the change in the altitude, in feet, of the hot air balloon each minute? Round your answer to the nearest foot.

1231 *feet*

**Notes on Scoring**

This response earns no credit (0 points) because the student calculates the altitude of the hot air balloon and gives this as the answer instead of the change in altitude.

The student recognizes the distances between the launching point, hot air balloon, and boat form a right triangle, so trigonometry can be used to calculate the altitude. Since the angle of elevation and the adjacent side of the triangle are given and the opposite side of the triangle is the unknown, the tangent function should be used to calculate the altitude. The altitude is $400\tan(72) \approx 1231$.

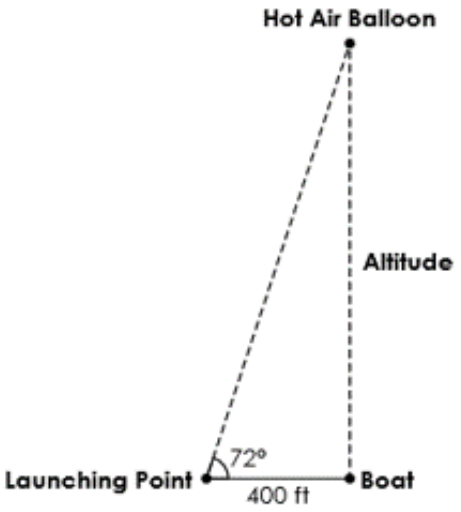
The student may stop here, overlooking the fact that the altitude has been calculated, not the change in altitude.

## Sample Response: 0 points

A hot air balloon is launched and drifts over a boat.

- The balloon's altitude increases at a constant rate.
- The balloon is directly above a boat 400 feet (ft) away from the launching point 5 minutes after it is launched.
- The angle of elevation is 72°.

A figure describing the situation is shown.




What is the change in the altitude, in feet, of the hot air balloon each minute? Round your answer to the nearest foot.

26 *feet*

**Notes on Scoring**

This response earns no credit (0 points) because the student incorrectly calculates the altitude of the balloon and uses that value to calculate the balloon's change in altitude.

The student recognizes the distances between the launching point, hot air balloon, and boat form a right triangle, so trigonometry can be used to calculate the altitude. Since the angle of elevation and the adjacent side of the triangle are given and the opposite side of the triangle is the unknown, the tangent function should be used to calculate the altitude. However, the student incorrectly applies the tangent formula as $\tan(\theta) = \frac{adjacent}{opposite}$ instead of $\tan(\theta) = \frac{opposite}{adjacent}$.

$$\begin{aligned} \tan\theta &= \frac{adjacent}{opposite} \\ \tan(72) &= \frac{400}{x} \\ x\tan(72) &= 400 \\ x &= \frac{400}{\tan(72)} \end{aligned}$$

To find the change in altitude of the hot air balloon per minute, the hot air balloon's altitude is divided by the amount of time it has been away from the launching point.

$$\frac{\frac{400}{\tan(72)}}{5} = x$$

$$26 \approx x$$

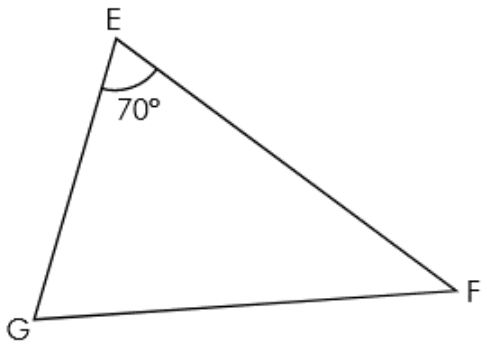
# Geometry
# Spring 2023 Item Release

# Question 30

# Question and Scoring Guidelines

## Question 30

Triangle EFG is shown, where EF = GF.




What is the measure of ∠ EFG?

Ⓐ 70°

Ⓑ 55°

Ⓒ 45°

Ⓓ 40°

**Points Possible:** 1

**Content Cluster:** Prove geometric theorems both formally and informally using a variety of methods.

**Content Standard:** Prove and apply theorems about triangles. *Theorems include but are not restricted to the following: measures of interior angles of a triangle sum to 180°; base angles of isosceles triangles are congruent; the segment joining midpoints of two sides of a triangle is parallel to the third side and half the length; the medians of a triangle meet at a point. (G.CO.10)*

**Depth of Knowledge:** Level 2

**Modeling and Reasoning:** No

# Scoring Guidelines

Rationale for Option A: This is incorrect. The student may not recognize that $\mathrm{EF} = \mathrm{GF}$ and incorrectly assume that $\mathrm{EG} = \mathrm{GF}$. Thus, $\angle\mathrm{EFG}$ is congruent to $\angle\mathrm{GEF}$ which is 70°.

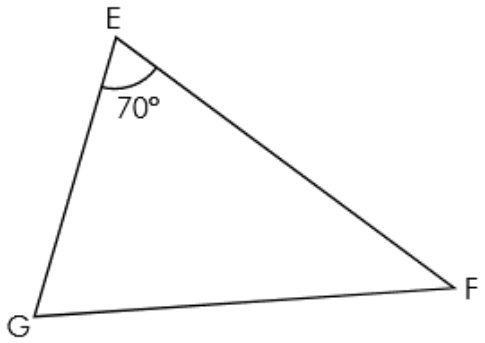
Rationale for Option B: This is incorrect. The student may assume that $\angle\mathrm{GEF}$ is the vertex angle and find the measure of $\angle\mathrm{EFG}$ as $(180° - 70°) \div 2 = 55°$.

Rationale for Option C: This is incorrect. The student may assume that $\angle\mathrm{GEF}$ is the apex angle and that the base angles are complementary. Thus, the measure of $\angle\mathrm{EFG}$ would be $90° \div 2 = 45°$.

Rationale for Option D: **Key** – The student correctly recognizes that the base angles of an isosceles triangle are congruent and therefore $\angle\mathrm{GEF} = \angle\mathrm{EFG} = 70°$. The student then calculates $\angle\mathrm{EFG}$ as $180° - (70° + 70°) = 40°$.

**Sample Response: 1 point**

Triangle EFG is shown, where EF = GF.




What is the measure of ∠EFG?

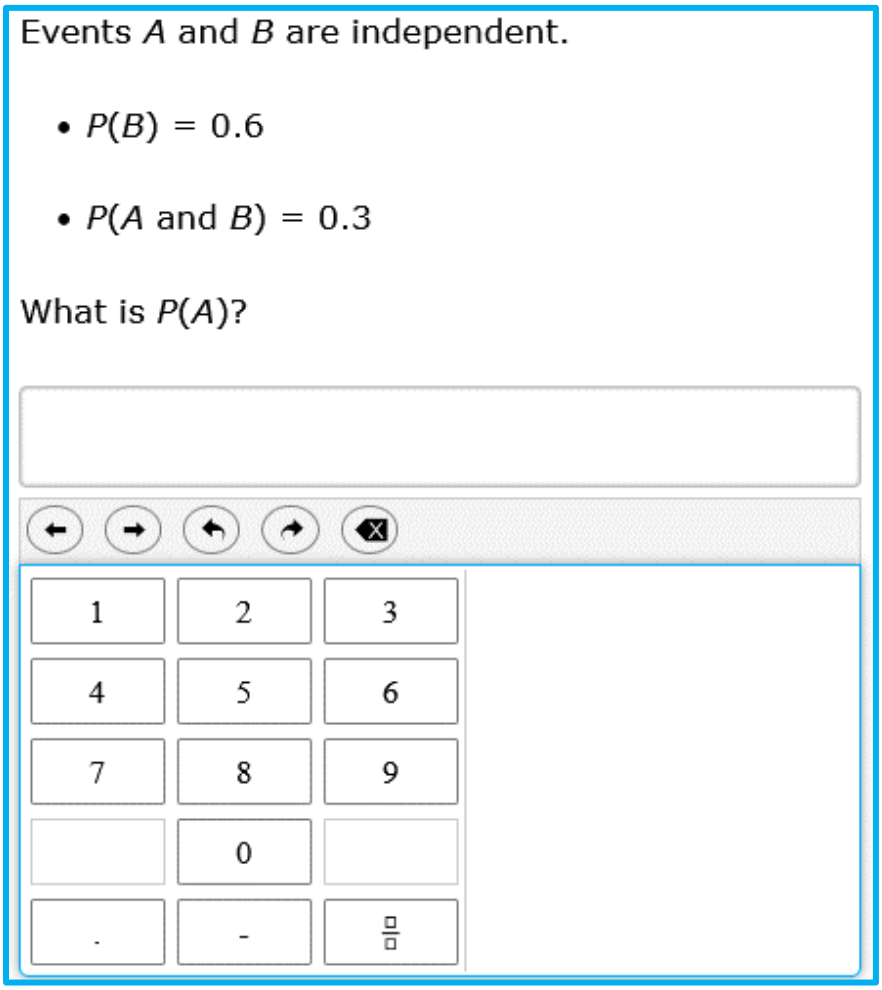
- Ⓐ 70°
- Ⓑ 55°
- Ⓒ 45°
- ● 40°

# Geometry
# Spring 2023 Item Release

# Question 31

# Question and Scoring Guidelines

# Question 31




**Points Possible:** 1

**Content Cluster:** Understand independence and conditional probability, and use them to interpret data.

**Content Standard:** Understand that two events A and B are independent if and only if the probability of A and B occurring together is the product of their probabilities, and use this characterization to determine if they are independent.★ *(S.CP.2)*

**Depth of Knowledge:** Level 2

**Modeling and Reasoning:** No

# Scoring Guidelines

For full credit (1 point), the student's response satisfies the bullet below.

- The student correctly determines that $P(A) = 0.5$, providing evidence of the ability to understand that the probability of two independent events happening is the product of each of their probabilities as a single event.

Exemplar Response

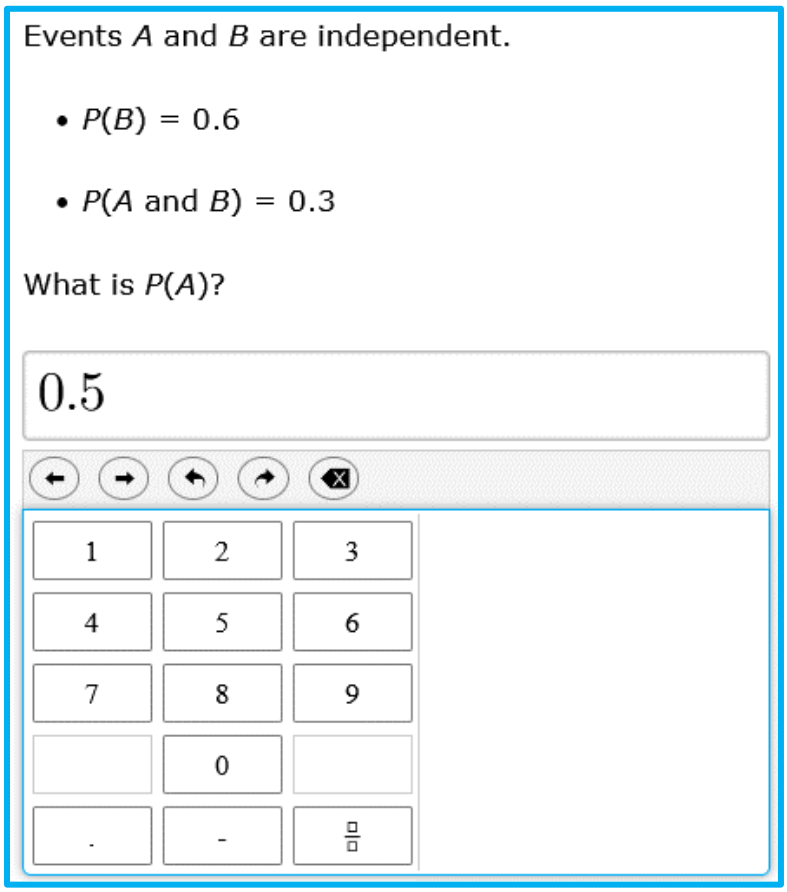
- 0.5
  or
- any equivalent value

# Geometry
# Spring 2023 Item Release

# Question 31

# Sample Responses

**Sample Response: 1 point**



**Notes on Scoring**

This response earns full credit (1 point) because the student correctly identifies the probability of *A*, given the probability of *A* and *B* and that they are independent events.

The student understands that since event *A* and event *B* are independent, the probability of *A* and *B* occurring, *P*(*A* and *B*), is equal to the product of the individual probabilities. The student may substitute values for *P*(*A* and *B*) and *P(*B) into the equation and solve for *P*(*A*).

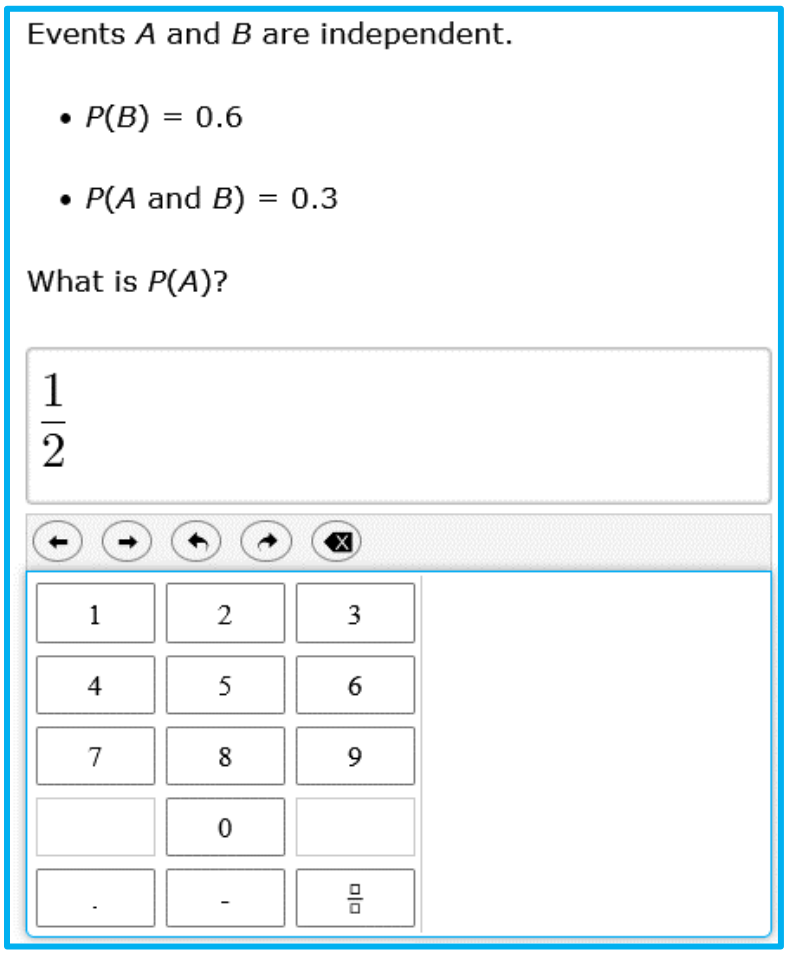
$$P(A \text{ and } B) = P(A) \cdot P(B)$$

$$0.3 = P(A) \cdot 0.6$$

$$\frac{0.3}{0.6} = P(A)$$

$$0.5 = P(A)$$

**Sample Response: 1 point**



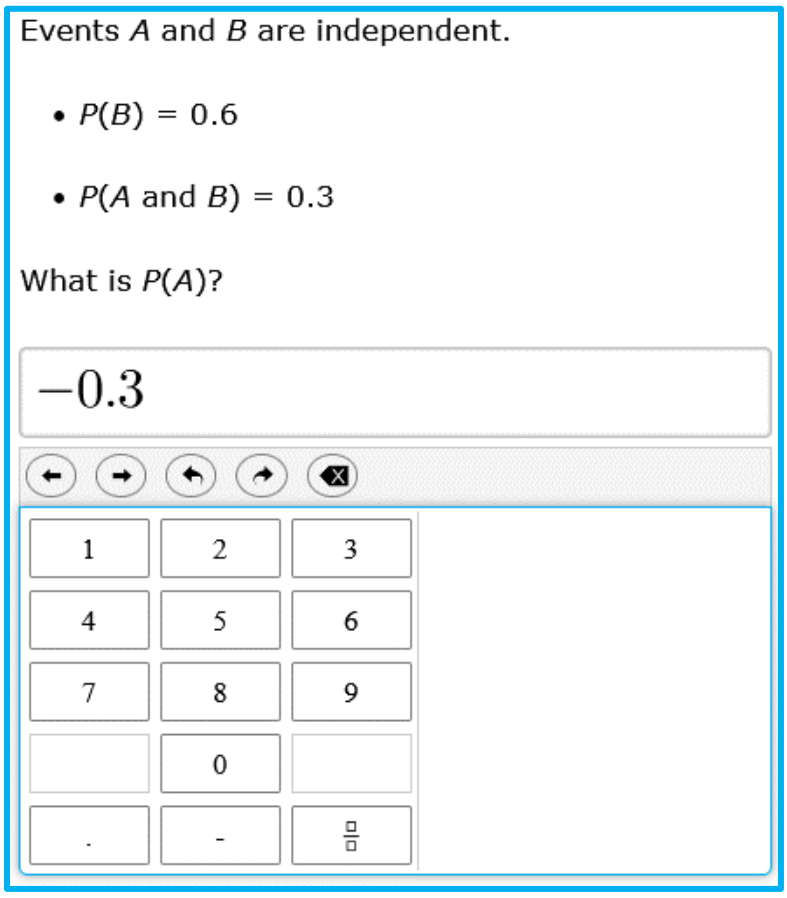
**Notes on Scoring**

This response earns full credit (1 point) because the student correctly identifies the probability of *A*, given the probability of *A* and *B* and that they are independent events.

The student understands that since event *A* and event *B* are independent, the probability of *A* and *B* occurring, P(*A* and *B*), is equal to the product of the individual probabilities.

$$
\begin{aligned}
P(A \text{ and } B) &= P(A) \cdot P(B) \\
0.3 &= P(A) \cdot 0.6 \\
\frac{0.3}{0.6} &= P(A) \\
0.5 &= P(A) \\
\frac{1}{2} &= P(A)
\end{aligned}
$$

**Sample Response: 0 points**



**Notes on Scoring**

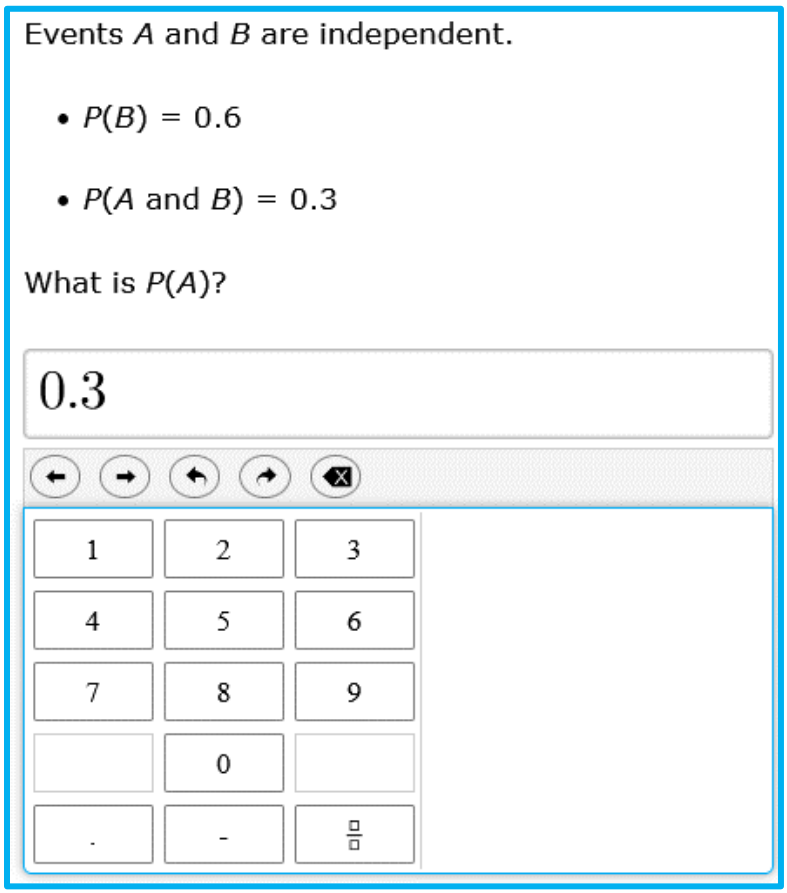
This response earns no credit (0 points) because the student incorrectly calculates the probability of *A*, given the probability of *A* and *B* and that they are independent events.

The student may think that since event *A* and event *B* are independent, the probability of *A* and *B* occurring, *P*(*A* and *B*), is equal to the sum of the individual probabilities instead of the product of the individual probabilities.

$$\begin{aligned} P(A \text{ and } B) &= P(A) + P(B) \\ 0.3 &= P(A) + 0.6 \\ -0.3 &= P(A) \end{aligned}$$

The student may not recognize that the probability of an event cannot be negative.

## Sample Response: 0 points




**Notes on Scoring**

This response earns no credit (0 points) because the student incorrectly calculates the probability of *A*, given the probability of *A* and *B* and that they are independent events.

The student may think that that since event *A* and event *B* are independent, the probability of *A* and *B* occurring, *P*(*A* and *B*), is equal to the sum of the individual probabilities instead of the product of the individual probabilities. The student calculates the *P*(*A*) as $-0.3$.

The student may then recognize that the probability of an event cannot be negative and drops the negative sign to obtain *P*(*A*) = 0.3.

# Geometry
# Spring 2023 Item Release

# Question 36

# Question and Scoring Guidelines

# Question 36

This item has **two** parts.

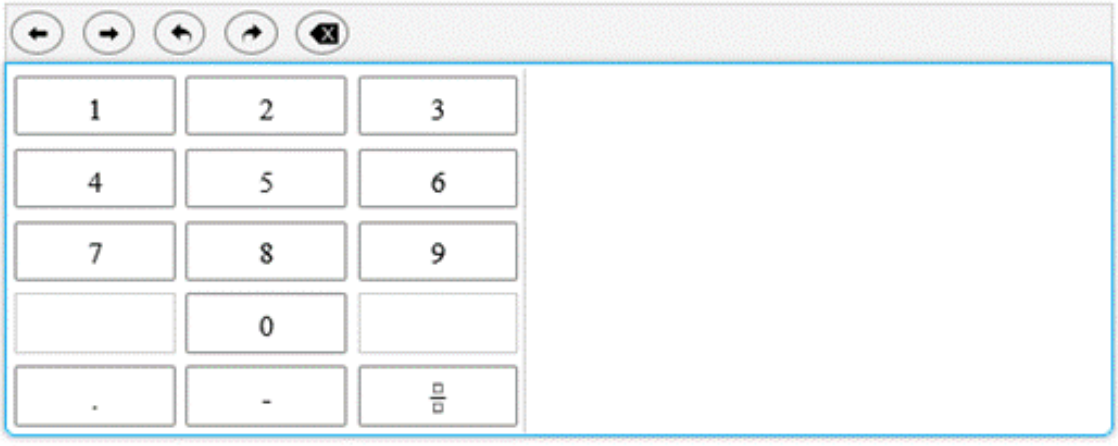
A high school teacher surveyed 700 junior and senior students to see whether they preferred biology or chemistry.

- Out of all the juniors surveyed, 35% said they preferred chemistry.
- Out of all the students surveyed, 56% said they preferred biology.

**Part A.** Complete the two-way table that shows how many junior and senior students preferred each science course.

| | Biology | Chemistry | Total |
|---|---|---|---|
| **Juniors** | | | 340 |
| **Seniors** | | | |
| **Total** | | | 700 |

**Part B.** What is the **exact** probability that a student is a senior given that the student preferred biology?



**Points Possible:** 2

**Content Cluster:** Understand independence and conditional probability, and use them to interpret data.

**Content Standard:** Construct and interpret two-way frequency tables of data when two categories are associated with each object being classified. Use the two-way table as a sample space to decide if events are independent and to approximate conditional probabilities. *For example, collect data from a random sample of students in your school on their favorite subject among math, science, and English. Estimate the probability that a randomly selected student from your school will favor science given that the student is in tenth grade. Do the same for other subjects and compare the results.★ (S.CP.4)*

**Depth of Knowledge:** Level 2

**Modeling and Reasoning:** Yes

# Scoring Guidelines

For 2 points, the student's response satisfies both bullets below.

For 1 point, the student's response satisfies one of the bullets below.

- The student correctly completes the two-way table, providing evidence of the ability to construct two-way tables of data.
- The student correctly determines the probability based on the information given or the table in Part A, providing evidence of the ability to use two-way tables to find conditional probabilities.

Exemplar Response: 2 points

- Part A:

| | Biology | Chemistry | Total |
|---|---|---|---|
| **Juniors** | 221 | 119 | 340 |
| **Seniors** | 171 | 189 | 360 |
| **Total** | 392 | 308 | 700 |

- Part B: $\frac{171}{392}$, or any equivalent value

Exemplar Response: 1 point

- Part A:

| | Biology | Chemistry | Total |
|---|---|---|---|
| **Juniors** | 102 | 238 | 340 |
| **Seniors** | 290 | 70 | 360 |
| **Total** | 392 | 308 | 700 |

- Part B: $\frac{290}{392}$, or any equivalent value

Note: The student will receive credit for correctly determining the ratio for a conditional probability, based on an incorrectly completed table in Part A, as long as the value is a valid probability between 0 and 1.

# Geometry
# Spring 2023 Item Release

# Question 36

# Sample Responses

## Sample Response: 2 points

This item has **two** parts.

A high school teacher surveyed 700 junior and senior students to see whether they preferred biology or chemistry.

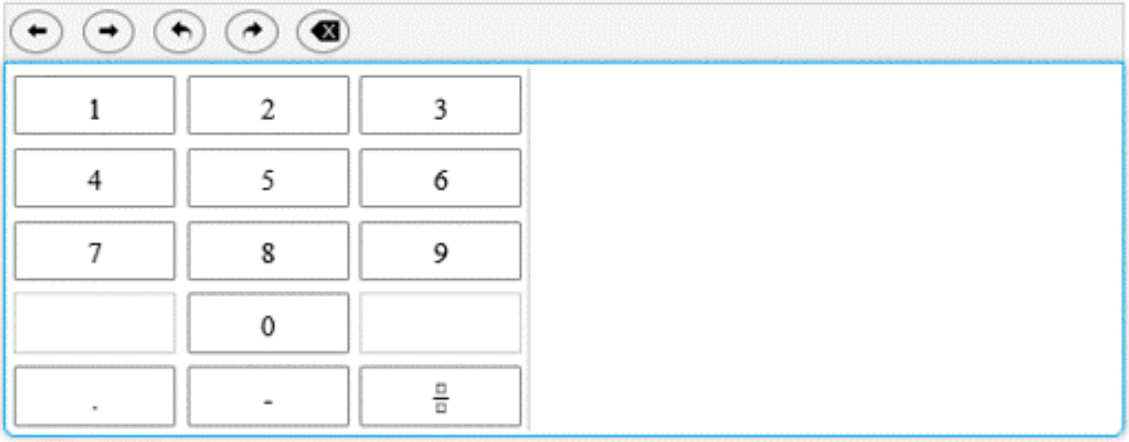
- Out of all the juniors surveyed, 35% said they preferred chemistry.
- Out of all the students surveyed, 56% said they preferred biology.

**Part A.** Complete the two-way table that shows how many junior and senior students preferred each science course.

| | Biology | Chemistry | Total |
|---|---|---|---|
| **Juniors** | 221 | 119 | 340 |
| **Seniors** | 171 | 189 | 360 |
| **Total** | 392 | 308 | 700 |

**Part B.** What is the **exact** probability that a student is a senior given that the student preferred biology?

$\frac{171}{392}$



**Notes on Scoring**

This response earns full credit (2 points) because the student completes the two-way table in Part A using the 2 bullets of information and then the information in the other cells of the table. In Part B, the student correctly answers the exact probability that a student is a senior given that the student preferred biology.

First, the information in the bullets is used to fill in missing information in the table.

> Since 35% of the juniors surveyed preferred chemistry, the number of juniors who preferred chemistry is calculated by multiplying the total number of juniors surveyed, 340, by 35% to get 119.
>
> Since 56% of the total number of students surveyed preferred biology, the total number of students surveyed who preferred biology is calculated by taking the total number of students surveyed, 700, and multiplying by 56% to get 392.

The missing cells remaining in the table can be found using the information already in the table.

> Since the total number of juniors surveyed is 340 and the number of juniors who preferred chemistry is 119, 119 can be subtracted from 340 to find the number of juniors who preferred biology is 221.
>
> Similarly, the total number of seniors surveyed who preferred biology can be calculated by subtracting the number of juniors who preferred biology, 221, from the total number of students surveyed who preferred biology, 392, to get 171.

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**Notes on Scoring**

To find the total number of students surveyed who preferred chemistry, the total number of students surveyed who preferred biology, 392, is subtracted from the total number of students surveyed, 700, to get 308.

Next, the number of juniors who preferred chemistry, 119, is subtracted from the total number of students surveyed who preferred chemistry, 308, to see that the number of seniors who preferred chemistry is 189.

Lastly, the total number of seniors surveyed needs to be calculated by adding the number of seniors who preferred biology, 171, and the number of seniors who preferred chemistry, 189, to get 360.

Now the student can calculate the answer to Part B. The probability that a student surveyed is a senior given that the student surveyed preferred biology can be calculated using the formula

$P(A|B) = \frac{P(A \text{ and } B)}{P(B)}$ where event *A* is that the student surveyed is a senior and event *B* is that the student surveyed preferred biology. Since *P*(student is a senior and student preferred biology) = $\frac{171}{700}$ and *P*(the student preferred biology) = $\frac{392}{700}$, the probability that a student surveyed is a senior given that the student surveyed preferred biology is $\frac{171}{392}$.

## Sample Response: 1 point

This item has **two** parts.

A high school teacher surveyed 700 junior and senior students to see whether they preferred biology or chemistry.

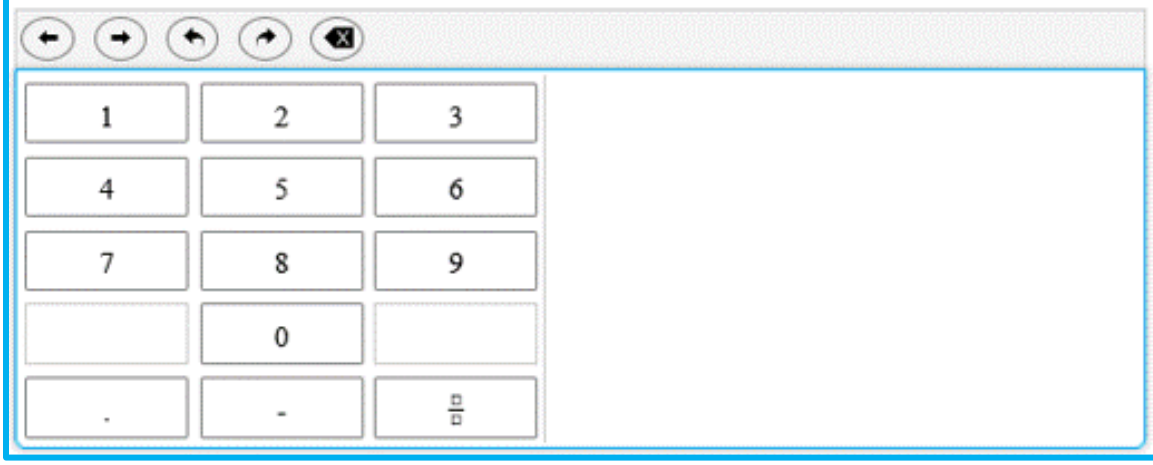
- Out of all the juniors surveyed, 35% said they preferred chemistry.
- Out of all the students surveyed, 56% said they preferred biology.

**Part A.** Complete the two-way table that shows how many junior and senior students preferred each science course.

| | Biology | Chemistry | Total |
|---|---|---|---|
| **Juniors** | 221 | 119 | 340 |
| **Seniors** | 171 | 243 | 360 |
| **Total** | 392 | 362 | 700 |

**Part B.** What is the **exact** probability that a student is a senior given that the student preferred biology?

$\frac{171}{392}$



**Notes on Scoring**

This response earns partial credit (1 point) because the student incorrectly completes the Chemistry column in the two-way table in Part A, but correctly calculates the probability that a student is a senior given that the student preferred biology.

When the student attempts to calculate the number of seniors who preferred chemistry, they may incorrectly subtract 117 from 360 instead of 171, resulting in 243 seniors who prefer chemistry instead of 189. The student may then add 119 + 243 to get the total number of students who prefer chemistry as 362. The student may not check their work though, because while the column entries add to the column totals, the row entries do not add to the row totals for the Seniors row ($171 + 243 \neq 360$) or the Total row $392 + 362 \neq 700$).

Now the student can calculate the answer to Part B. The probability that a student surveyed is a senior given that the student surveyed preferred biology can be calculated using the formula $P(A|B) = \frac{P(A \text{ and } B)}{P(B)}$ where event *A* is that the student surveyed is a senior and event *B* is that the student surveyed preferred biology. Since *P*(student is a senior and student preferred biology) = $\frac{171}{700}$ and *P*(the student preferred biology) = $\frac{392}{700}$, the probability that a student surveyed is a senior given that the student surveyed preferred biology is $\frac{171}{392}$.

## Sample Response: 1 point

This item has **two** parts.

A high school teacher surveyed 700 junior and senior students to see whether they preferred biology or chemistry.

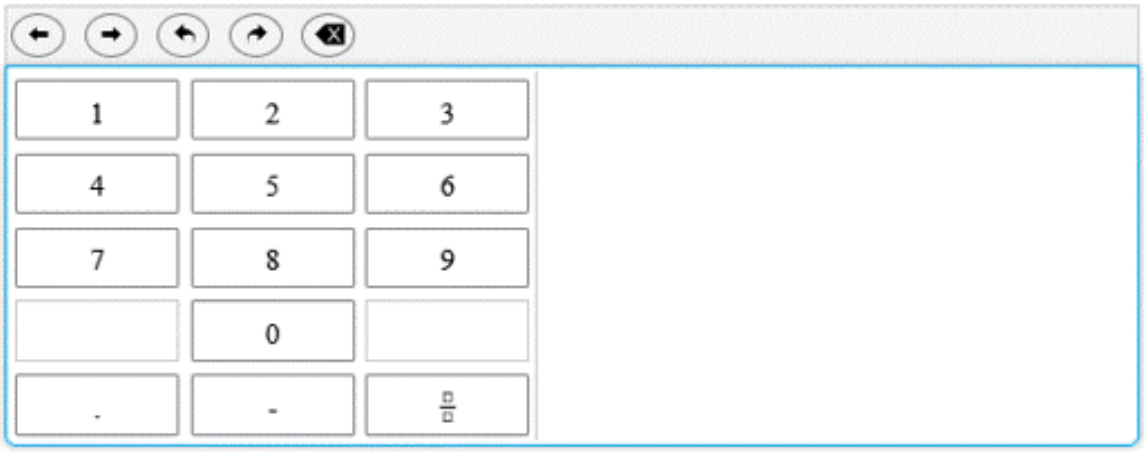
- Out of all the juniors surveyed, 35% said they preferred chemistry.
- Out of all the students surveyed, 56% said they preferred biology.

**Part A.** Complete the two-way table that shows how many junior and senior students preferred each science course.

| | Biology | Chemistry | Total |
|---|---|---|---|
| **Juniors** | 221 | 119 | 340 |
| **Seniors** | 171 | 189 | 360 |
| **Total** | 392 | 308 | 700 |

**Part B.** What is the **exact** probability that a student is a senior given that the student preferred biology?

$\frac{171}{700}$



**Notes on Scoring**

This response earns partial credit (1 point) because the student completes the two-way table in Part A using the 2 bullets of information and then the information in the other cells of the table. In Part B, the student incorrectly answers the exact probability that a student is a senior given that the student preferred biology.

First, the information in the bullets is used to fill in missing information in the table.

> Since 35% of the juniors surveyed preferred chemistry, the number of juniors who preferred chemistry is calculated by multiplying the total number of juniors surveyed, 340, by 35% to get 119.
>
> Since 56% of the total number of students surveyed preferred biology, the total number of students surveyed who preferred biology is calculated by taking the total number of students surveyed, 700, and multiplying by 56% to get 392.

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**Notes on Scoring**

The missing cells remaining in the table can be found using the information already in the table.

Since the total number of juniors surveyed is 340 and the number of juniors who preferred chemistry is 119, 119 can be subtracted from 340 to find the number of juniors who preferred biology is 221.

Similarly, the total number of seniors surveyed who preferred biology can be calculated by subtracting the number of juniors who preferred biology, 221, from the total number of students surveyed who preferred biology, 392, to get 171.

To find the total number of students surveyed who preferred chemistry, the total number of students surveyed who preferred biology, 392, is subtracted from the total number of students surveyed, 700, to get 308.

Next, the number of juniors who preferred chemistry, 119, is subtracted from the total number of students surveyed who preferred chemistry, 308, to see that the number of seniors who preferred chemistry is 189.

Lastly, the total number of seniors surveyed needs to be calculated by adding the number of seniors who preferred biology, 171, and the number of seniors who preferred chemistry, 189, to get 360.

Now the student can calculate the answer to Part B. The student may misinterpret the probability that a student surveyed is a senior given that the student surveyed preferred biology as the probability the student surveyed is a senior and preferred biology, and thus calculates the probability as the number of students surveyed who are a senior and preferred biology, 171, divided by the total number of students surveyed, 700.

## Sample Response: 0 points

This item has **two** parts.

A high school teacher surveyed 700 junior and senior students to see whether they preferred biology or chemistry.

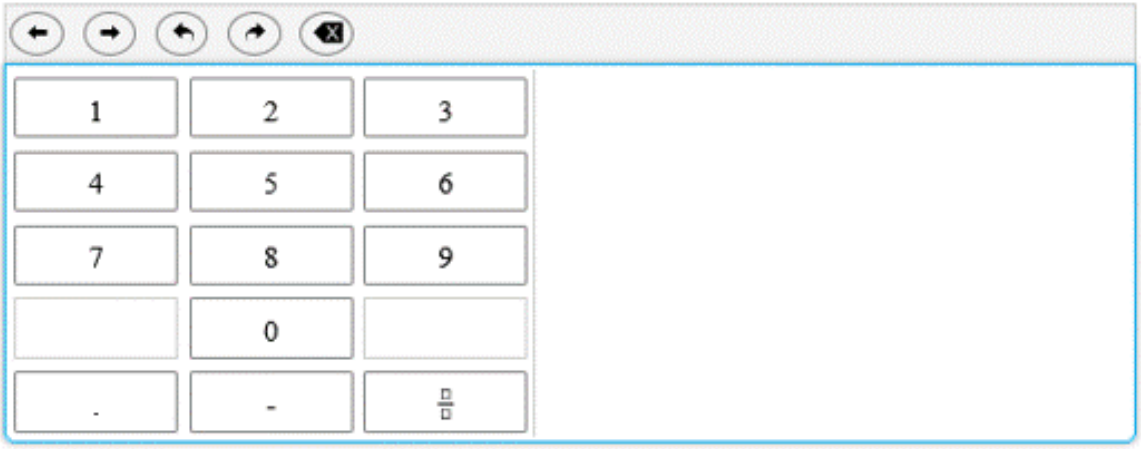
- Out of all the juniors surveyed, 35% said they preferred chemistry.
- Out of all the students surveyed, 56% said they preferred biology.

**Part A.** Complete the two-way table that shows how many junior and senior students preferred each science course.

| | Biology | Chemistry | Total |
|---|---|---|---|
| **Juniors** | 120 | 220 | 340 |
| **Seniors** | 130 | 230 | 360 |
| **Total** | 250 | 450 | 700 |

**Part B.** What is the **exact** probability that a student is a senior given that the student preferred biology?

$\frac{360}{700}$



**Notes on Scoring**

This response earns no credit (0 points) because the student incorrectly completes the two-way table in Part A using the 2 bullets of information and then the information in the other cells of the table. In Part B, the student incorrectly answers the exact probability that a student is a senior given that the student preferred biology.

First, the information in the bullets is used to fill in the missing information in the table. The number of juniors who preferred chemistry is calculated by multiplying the total number of juniors, 340, by 35% to get 119 since 35% of the juniors surveyed preferred chemistry. The student may round 119 to the nearest ten to get 120 and place the number in the first empty cell in the table, juniors who prefer biology, instead of the cell for juniors who prefer chemistry.

A few of the cells in the table can be found using information already in the table. Since the juniors who preferred biology is 120 and the total number of juniors surveyed is 340, 120 can be subtracted from 340 to find the number of juniors who prefer chemistry is 220. The total number of juniors surveyed, 340, can be subtracted from the total number of students surveyed, 700, to find the total number of seniors surveyed is 360.

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**Notes on Scoring**

The student may think that 35% of seniors preferred biology as well and calculates the number of seniors who preferred biology by multiplying the total number of seniors surveyed, 360, by 35% to get 126. The student may round 126 to the nearest ten to get 130.

Since the total number of seniors who preferred biology is 130 and the total number of seniors surveyed is 360, 130 can be subtracted from 360 to find the number of seniors who preferred chemistry is 230. To find the total number of students who preferred biology, the juniors who preferred biology, 120, can be added to the seniors who preferred biology, 130, to get 250.

Similarly, to find the total number of students surveyed who preferred chemistry, the juniors who preferred chemistry, 220, needs to be added to the seniors who preferred chemistry, 230, to get 450.

Now the student can calculate the answer to Part B. The student may misinterpret the probability that a student surveyed is a senior given that the student surveyed preferred biology as the probability the student surveyed is a senior, and thus calculates the probability based on their table in Part A as the number of students surveyed who are a senior, 360, divided by the total number of students surveyed, 700.

## Sample Response: 0 points

This item has **two** parts.

A high school teacher surveyed 700 junior and senior students to see whether they preferred biology or chemistry.

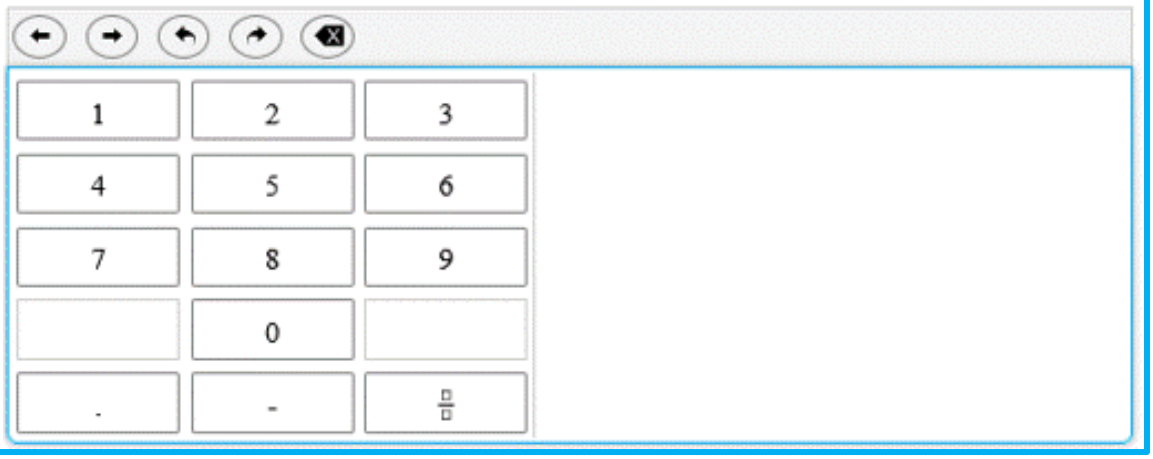
- Out of all the juniors surveyed, 35% said they preferred chemistry.
- Out of all the students surveyed, 56% said they preferred biology.

**Part A.** Complete the two-way table that shows how many junior and senior students preferred each science course.

| | Biology | Chemistry | Total |
|---|---|---|---|
| **Juniors** | 120 | 220 | 340 |
| **Seniors** | 140 | 220 | 360 |
| **Total** | 260 | 440 | 700 |

**Part B.** What is the **exact** probability that a student is a senior given that the student preferred biology?

$\frac{140}{360}$



**Notes on Scoring**

This response earns no credit (0 points) because the student incorrectly completes the two-way table in Part A using the 2 bullets of information and then the information in the other cells of the table. In Part B, the student incorrectly answers the exact probability that a student is a senior given that the student preferred biology.

First, the information in the bullets is used to fill in the missing information in the table. Since 35% of the juniors surveyed preferred chemistry, the number of juniors who preferred chemistry is calculated by multiplying the total number of juniors, 340, by 35% to get 119. The student may round 119 to the nearest ten to get 120 and place the number in the first empty cell in the table, juniors who prefer biology, instead of the cell for juniors who prefer chemistry in the table.

A few of the cells in the table can be found using information already in the table. Since the juniors who preferred biology is 120 and the total number of juniors surveyed is 340, 120 can be subtracted from 340 to find the number of juniors who preferred chemistry as 220. The total number of juniors surveyed, 340, can be subtracted from the total number of students surveyed, 700, to find the total number of seniors surveyed is 360.

The student may think that 35% of seniors preferred biology as well and multiply the total number of seniors surveyed, 360, by 35% to get 126, and then round 126 up to get 140.

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**Notes on Scoring**

Since the total number of seniors who preferred biology is 140 and the total number of seniors surveyed is 360, 140 can be subtracted from 360 to find the number of seniors who preferred chemistry is 220. To find the total number of students who preferred biology, the juniors who preferred biology, 120, needs to be added to the seniors who preferred biology, 140, to get 260.

Similarly, to find the total number of students who preferred chemistry, the juniors who preferred chemistry, 220, needs to be added to the seniors who preferred chemistry, 220, to get 440.

Now the student can calculate the answer to Part B. The student may misinterpret the probability that a student surveyed is a senior given that the student surveyed preferred biology as the probability the student surveyed is a senior, and thus calculates the probability based on their table in Part A as the number of students surveyed who are a senior, 140, divided by the total number of seniors surveyed, 360.

# Geometry
# Spring 2023 Item Release

# Question 37

# Question and Scoring Guidelines

# Question 37

Select the boxes to identify whether each statement is always, sometimes, or never true.

| | Always | Sometimes | Never |
|---|---|---|---|
| **A square has four right angles.** | ☐ | ☐ | ☐ |
| **A square has two pairs of parallel sides.** | ☐ | ☐ | ☐ |
| **A parallelogram has four congruent sides.** | ☐ | ☐ | ☐ |
| **A parallelogram has four right angles.** | ☐ | ☐ | ☐ |
| **A parallelogram has two pairs of congruent opposite sides.** | ☐ | ☐ | ☐ |

**Points Possible:** 1

**Content Cluster:** Classify and analyze geometric figures.

**Content Standard:** Classify two-dimensional figures in a hierarchy based on properties. *(G.CO.14)*

**Depth of Knowledge:** Level 2

**Modeling and Reasoning:** Yes

# Scoring Guidelines

For full credit (1 point), the student's response satisfies the bullet below.

- The student identifies all of the statements correctly as always, sometimes, or never true, providing evidence of the ability to classify two-dimensional objects based on a hierarchy.

Exemplar Response

| | Always | Sometimes | Never |
|---|---|---|---|
| **A square has four right angles.** | ☑ | ☐ | ☐ |
| **A square has two pairs of parallel sides.** | ☑ | ☐ | ☐ |
| **A parallelogram has four congruent sides.** | ☐ | ☑ | ☐ |
| **A parallelogram has four right angles.** | ☐ | ☑ | ☐ |
| **A parallelogram has two pairs of congruent opposite sides.** | ☑ | ☐ | ☐ |

# Geometry
# Spring 2023 Item Release

# Question 37

# Sample Responses

## Sample Response: 1 point

Select the boxes to identify whether each statement is always, sometimes, or never true.

| | Always | Sometimes | Never |
|---|---|---|---|
| **A square has four right angles.** | ☑ | ☐ | ☐ |
| **A square has two pairs of parallel sides.** | ☑ | ☐ | ☐ |
| **A parallelogram has four congruent sides.** | ☐ | ☑ | ☐ |
| **A parallelogram has four right angles.** | ☐ | ☑ | ☐ |
| **A parallelogram has two pairs of congruent opposite sides.** | ☑ | ☐ | ☐ |

**Notes on Scoring**

This response earns full credit (1 point) because the student correctly identifies whether each statement is always, sometimes, or never true.

The student identifies properties of a square and parallelogram by selecting a square always has four right angles, a square always has two pairs of parallel sides, a parallelogram sometimes has four congruent sides (only when it is also a rhombus or a square), a parallelogram sometimes has four right angles (only when it is also a rectangle or a square), and a parallelogram always has two pairs of congruent opposite sides.

## Sample Response: 0 points

Select the boxes to identify whether each statement is always, sometimes, or never true.

| | Always | Sometimes | Never |
|---|---|---|---|
| **A square has four right angles.** | ☑ | ☐ | ☐ |
| **A square has two pairs of parallel sides.** | ☑ | ☐ | ☐ |
| **A parallelogram has four congruent sides.** | ☐ | ☑ | ☐ |
| **A parallelogram has four right angles.** | ☐ | ☑ | ☐ |
| **A parallelogram has two pairs of congruent opposite sides.** | ☐ | ☑ | ☐ |

**Notes on Scoring**

This response earns no credit (0 points) because the student incorrectly identifies one of the statements.

The student identifies that a square always has four right angles and two pairs of parallel sides and that a parallelogram sometimes has four congruent sides and sometimes has four right angles, but may incorrectly think that a parallelogram only sometimes has two pairs of congruent opposite sides not realizing a parallelogram always has two pairs of congruent opposite sides.

## Sample Response: 0 points

Select the boxes to identify whether each statement is always, sometimes, or never true.

| | Always | Sometimes | Never |
|---|---|---|---|
| **A square has four right angles.** | ☑ | ☐ | ☐ |
| **A square has two pairs of parallel sides.** | ☑ | ☐ | ☐ |
| **A parallelogram has four congruent sides.** | ☐ | ☑ | ☐ |
| **A parallelogram has four right angles.** | ☐ | ☐ | ☑ |
| **A parallelogram has two pairs of congruent opposite sides.** | ☐ | ☑ | ☐ |

**Notes on Scoring**

This response earns no credit (0 points) because the student incorrectly identifies two of the statements.

The student may identify that a square always has four right angles and two pairs of parallel sides and that a parallelogram sometimes has four congruent sides. The student may incorrectly think that a parallelogram never has four right angles, not realizing a rectangle and a square are defined as parallelograms that have four right angles. The student may also incorrectly think a parallelogram sometimes has two pairs of congruent opposite sides, not realizing a parallelogram always has two pairs of congruent opposite sides.

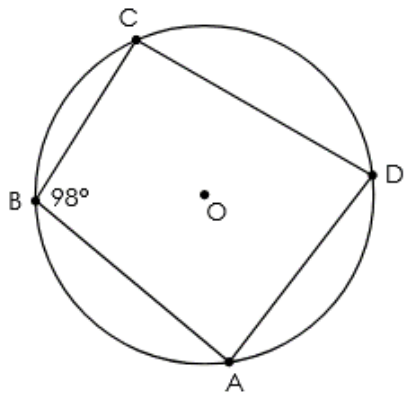
# Geometry
# Spring 2023 Item Release

# Question 46

# Question and Scoring Guidelines

# Question 46

Quadrilateral ABCD is inscribed in circle O, as shown.




What is the measure, in degrees, of $\angle CDA$ ?

degrees

**Points Possible:** 1

**Content Cluster:** Understand and apply theorems about circles.

**Content Standard:** Construct the inscribed and circumscribed circles of a triangle; prove and apply the property that opposite angles are supplementary for a quadrilateral inscribed in a circle. *(G.C.3)*

**Depth of Knowledge:** Level 1

**Modeling and Reasoning:** No

# Scoring Guidelines

For full credit (1 point), the student's response satisfies the bullet below.

- The student correctly enters 82, providing evidence of being able to apply the fact that opposite angles of a quadrilateral inscribed in a circle are supplementary.

Exemplar Response

- 82 degrees
  or
- any equivalent value

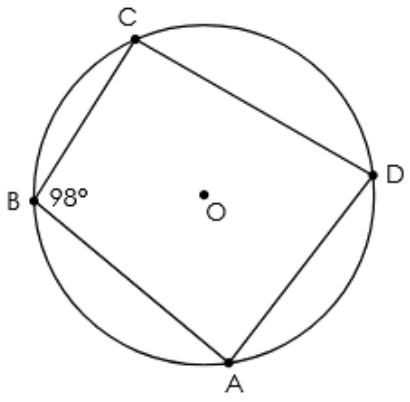
# Geometry
# Spring 2023 Item Release

# Question 46

# Sample Responses

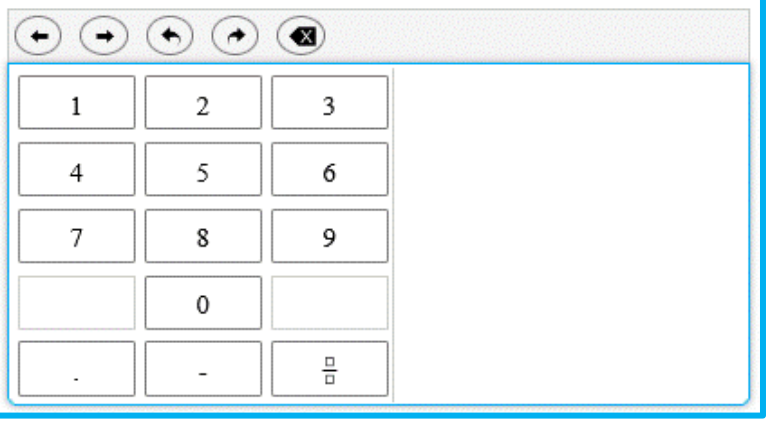
## Sample Response: 1 point

Quadrilateral ABCD is inscribed in circle O, as shown.




What is the measure, in degrees, of $\angle CDA$ ?

82 degrees



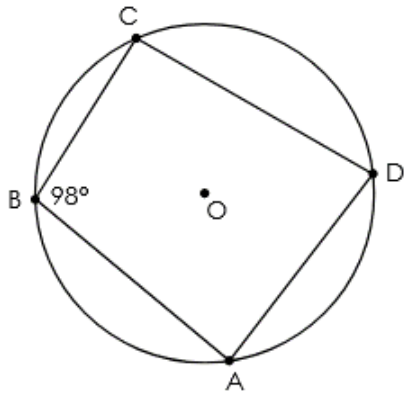

**Notes on Scoring**

This response earns full credit (1 point) because the student correctly calculates the measure of angle CDA.

The student recognizes that opposite angles of a quadrilateral inscribed in a circle are supplementary, meaning that the sum of their measures is 180 degrees. Since the measure of angle CBA is 98 degrees, the measure of angle CDA can be calculated as 180 – 98 = 82 degrees.

## Sample Response: 0 points

Quadrilateral ABCD is inscribed in circle O, as shown.




What is the measure, in degrees, of $\angle CDA$ ?

98 degrees

| | | |
|---|---|---|
| 1 | 2 | 3 |
| 4 | 5 | 6 |
| 7 | 8 | 9 |
| | 0 | |
| . | - | |

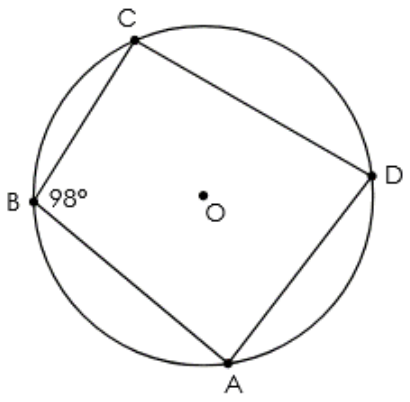
**Notes on Scoring**

This response earns no credit (0 points) because the student incorrectly calculates the measure of angle CDA.

The student may incorrectly think that opposite angles of a quadrilateral inscribed in a circle are congruent. Since the measure of angle CBA is 98 degrees and angle CDA is the angle opposite of angle CBA, the student determines the measure of angle CDA to be 98 degrees too.

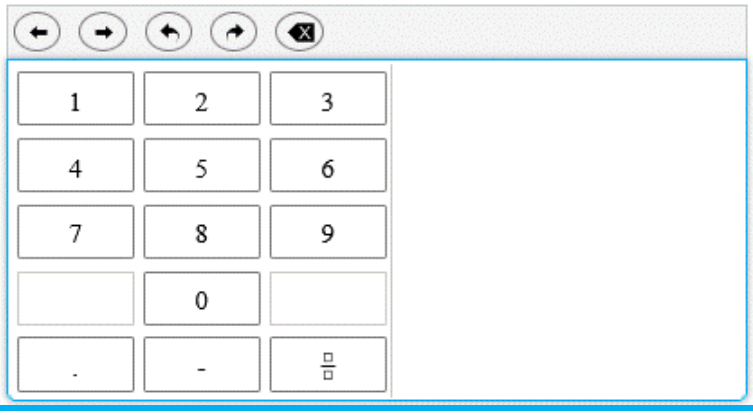
## Sample Response: 0 points

Quadrilateral ABCD is inscribed in circle O, as shown.




What is the measure, in degrees, of $\angle CDA$ ?

262 *degrees*




**Notes on Scoring**

This response earns no credit (0 points) because the student incorrectly identifies the measure of angle CDA.

The student may think that the sum of two opposite angles of a quadrilateral inscribed in a circle is 360 degrees and that since the measure of angle CBA is 98 degrees, the measure of angle CDA can be calculated as 360 – 98 = 262 degrees.

# Geometry
# Spring 2023 Item Release

# Question 47

# Question and Scoring Guidelines

# Question 47

What is the slope of a line that is perpendicular to the graph of $y = \frac{5}{4}x - \frac{3}{2}$ ?

(A) $-\frac{4}{5}$

(B) $-\frac{5}{4}$

(C) $\frac{4}{5}$

(D) $\frac{5}{4}$

**Points Possible:** 1

**Content Cluster:** Use coordinates to prove simple geometric theorems algebraically and to verify specific geometric statements.

**Content Standard:** Justify the slope criteria for parallel and perpendicular lines, and use them to solve geometric problems, e.g., find the equation of a line parallel or perpendicular to a given line that passes through a given point. *(G.GPE.5)*

**Depth of Knowledge:** Level 1

**Modeling and Reasoning:** No

# Scoring Guidelines

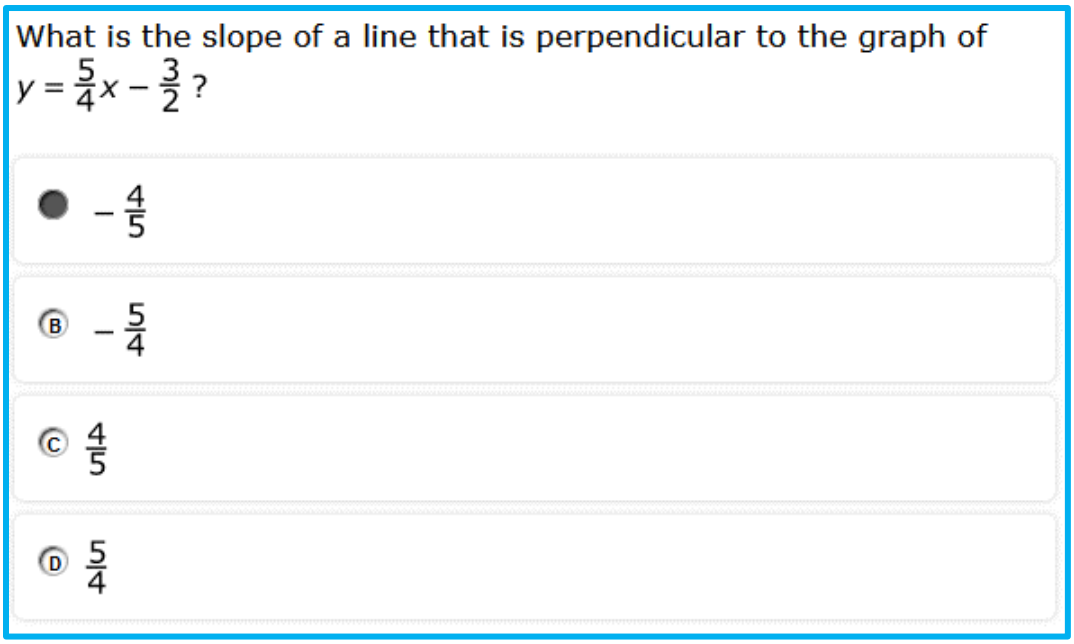
Rationale for Option A: **Key** – The student identifies the slope of the perpendicular line as $-\frac{4}{5}$ because it is the negative reciprocal of the slope, $\frac{5}{4}$.

Rationale for Option B: This is incorrect. The student may identify the slope of the given line as $\frac{5}{4}$ but may assume that the slope of a line perpendicular to the given line will be the negative of this slope.

Rationale for Option C: This is incorrect. The student may identify that the slope of the given line is $\frac{5}{4}$ but may think that the slope of a line perpendicular to the given line would have a slope that is the reciprocal of $\frac{5}{4}$ that is $\frac{4}{5}$.

Rationale for Option D: This is incorrect. The student may confuse the relationship between the slopes of perpendicular lines with the slopes of parallel lines and identifies a slope that is the same as the slope of the given line.

## Sample Response: 1 point



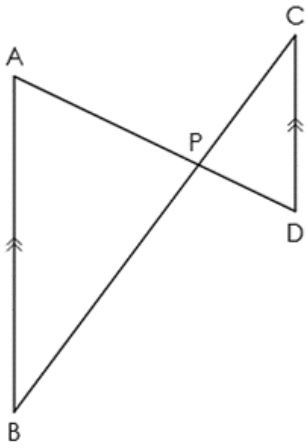
# Geometry
# Spring 2023 Item Release

# Question 48

# Question and Scoring Guidelines

# Question 48

A figure is shown, where $\overline{AB} \parallel \overline{CD}$.




Complete the statement to describe the relationship between the two triangles.

Triangle APB is [ ] to triangle DPC because triangle APB is a [ ] of triangle DPC.

**Drop Down Choices**

Triangle APB is [ ] to triangle DPC because triangle

- ✔
- similar
- not similar

APB is a [ ] of triangle DPC.

- ✔
- rotation and a dilation
- rotation and a reflection
- translation and a dilation
- reflection and a translation

**Points Possible:** 1

**Content Cluster:** Understand similarity in terms of similarity transformations.

**Content Standard:** Given two figures, use the definition of similarity in terms of similarity transformations to decide if they are similar; explain using similarity transformations the meaning of similarity for triangles as the equality of all corresponding pairs of angles and the proportionality of all corresponding pairs of sides. *(G.SRT.2)*

**Depth of Knowledge:** Level 2

**Modeling and Reasoning:** No

# Scoring Guidelines

For full credit (1 point), the student's response satisfies the bullet below.

- The student correctly identifies that triangle APB is similar to triangle DPC because triangle APB is a rotation and a dilation of triangle DPC, providing evidence of the ability to use the definition of similarity in terms of similarity transformations to decide whether they are similar.

Exemplar Response

- Triangle APB is **similar** to triangle DPC because triangle APB is a **rotation and a dilation** of triangle DPC.

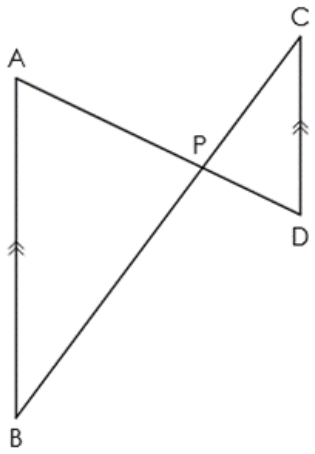
# Geometry
# Spring 2023 Item Release

# Question 48

# Sample Responses

## Sample Response: 1 point

A figure is shown, where $\overline{AB} \parallel \overline{CD}$.




Complete the statement to describe the relationship between the two triangles.

Triangle APB is [similar] to triangle DPC because triangle APB is a [rotation and a dilation] of triangle DPC.

**Notes on Scoring**

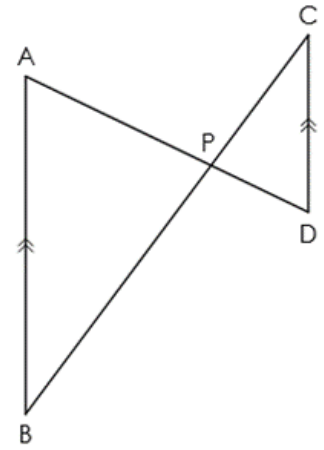
This response earns full credit (1 point) because the student correctly completes the statement to describe the relationship between the two triangles.

The student recognizes since $\overline{AB}$ is parallel to $\overline{CD}$, then angle ABP is congruent to angle DCP and angle BAP is congruent to angle CDP since alternate interior angles are congruent when a pair of parallel lines are cut by a transversal. Since these two pairs of corresponding angles are congruent, the triangles are similar by angle-angle similarity.

Since the two triangles are similar, one triangle can be mapped to the other with a series of transformations that includes a dilation. Since the options allow for only one additional transformation in addition to the dilation, the student recognizes that a rotation is needed to get vertices in one triangle in the same orientation as the vertices in the other triangle.

## Sample Response: 0 points

A figure is shown, where $\overline{AB} \parallel \overline{CD}$.




Complete the statement to describe the relationship between the two triangles.

Triangle APB is [similar] to triangle DPC because triangle APB is a [translation and a dilation] of triangle DPC.

**Notes on Scoring**

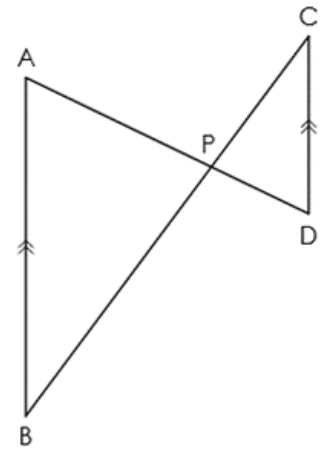
This response earns no credit (0 points) because the student incorrectly completes the statement to describe the relationship between the two triangles.

The student may recognize that the two triangles can be thought of as a pair of parallel lines cut by two transversals and use the facts about certain angles in such a case to determine that the two triangles are similar.

The student may also observe that since the triangles are different sizes that a dilation is needed to map one triangle onto the other. However, they may identify the other transformation that can be used to map one triangle onto the other as a translation to slide point A onto point C without realizing that the other 2 pairs of corresponding vertices will not also coincide with this transformation.

## Sample Response: 0 points

A figure is shown, where $\overline{AB} \parallel \overline{CD}$.




Complete the statement to describe the relationship between the two triangles.

Triangle APB is [similar] to triangle DPC because triangle APB is a [reflection and a translation] of triangle DPC.

**Notes on Scoring**

This response earns no credit (0 points) because the student incorrectly completes the statement to describe the relationship between the two triangles.

The student may recognize that the two triangles can be thought of as a pair of parallel lines cut by two transversals and use the facts about certain angles in such a case to determine that the two triangles are similar.

The student may think that a reflection across a line that goes through point P and is parallel to $\overline{AB}$ will result in angles APB and DPC coinciding and then thinking a translation up is needed so vertex B will coincide with vertex D, overlooking the fact that angles ABP and CDP are not congruent.

# Geometry
# Spring 2023 Item Release

# Question 49

# Question and Scoring Guidelines

# Question 49

A student flips a fair coin twice and is surprised the result is heads both times.

Which sentence correctly describes the relationship between the two events?

Ⓐ The two events are dependent because the result of the first flip changes the probability of the second flip being heads.

Ⓑ The two events are dependent because the result of the first flip does not change the probability of the second flip being heads.

Ⓒ The two events are independent because the result of the first flip changes the probability of the second flip being heads.

Ⓓ The two events are independent because the result of the first flip does not change the probability of the second flip being heads.

**Points Possible:** 1

**Content Cluster:** Understand independence and conditional probability, and use them to interpret data.

**Content Standard:** Recognize and explain the concepts of conditional probability and independence in everyday language and everyday situations. *For example, compare the chance of having lung cancer if you are a smoker with the chance of being a smoker if you have lung cancer.★ (S.CP.5)*

**Depth of Knowledge:** Level 2

**Modeling and Reasoning:** No

# Scoring Guidelines

Rationale for Option A: This is incorrect. The student may think that the result of the first flip affects the result of the second flip and then concludes that the events are dependent.

Rationale for Option B: This is incorrect. The student may recognize that the result of the first flip has no effect on the result of the second, but may confuse the definitions of dependent and independent events and conclude that the two events are dependent.

Rationale for Option C: This is incorrect. The student may recognize the events as independent but may incorrectly think that the result of the first flip has an effect on the second.

Rationale for Option D: **Key** – The student recognizes that the result of the first flip has no effect on the result of the second flip and concludes that this means the two events are independent.

## Sample Response: 1 point

A student flips a fair coin twice and is surprised the result is heads both times.

Which sentence correctly describes the relationship between the two events?

Ⓐ The two events are dependent because the result of the first flip changes the probability of the second flip being heads.

Ⓑ The two events are dependent because the result of the first flip does not change the probability of the second flip being heads.

Ⓒ The two events are independent because the result of the first flip changes the probability of the second flip being heads.

● The two events are independent because the result of the first flip does not change the probability of the second flip being heads.